When the Bookkeeping Breaks: Noise Thresholds and Failure Mechanisms of Adaptive Variational Berry-Phase Estimation

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Abstract

Adaptive variational quantum dynamics can compute quantized many-body Berry phases with circuits three orders of magnitude shallower than Trotterized adiabatic loops of matched noiseless accuracy, but only noiseless results have been reported. We characterize how much hardware noise this approach tolerates. On the dimerized Fermi–Hubbard chain (four sites, eight qubits, $U = 0$ and $U = 10$, both topological phases) we reproduce the noiseless results and then execute every circuit the method needs (the Hadamard-test readout and the 10^5–10^6 circuits per loop that measure the classical global-phase integrands) in a gate-level density-matrix simulation with depolarizing, relaxation, dephasing and readout errors scaled from a trapped-ion-class profile, and with genuine finite sampling. With a tolerance of 0.1π fixed in advance, the Berry phase stays accurate at 10^4 shots per circuit only below a two-qubit error of 9×10^{-5} ($U = 0$) and 8×10^{-6} ($U = 10$). The failure is a deterministic drift, not shot noise: about twenty shots per circuit suffice without gate noise. It originates in the classical bookkeeping. Time reversal cancels about 90% of the noise-induced dynamical-phase drift, but the residual dominates because adaptive growth makes the two halves of the loop inequivalent at the circuit level, and the geometric leg contracts generator by generator while the readout phase does not. Zero-noise extrapolation raises the threshold by more than a factor of 6.5. Without it, the compression pays off only against the matched-accuracy Trotter loop, whose threshold is 6.0–47$\times$ lower: the quantized phase tolerates a hundred times coarser Trotter step, and that single-circuit loop tolerates 1.5$\times$ ($U = 0$) and 15$\times$ ($U = 10$) more noise than the unmitigated variational method.

1 Introduction

Berry phases [Berry, 1984] classify one-dimensional topological matter: for a time-reversal-invariant chain threaded by a twist, the many-body Berry phase is quantized to 0 or π [Zak, 1989, Resta, 1994, Niu et al., 1985, Xiao et al., 2010, Hasan and Kane, 2010]. Evaluating it for interacting systems is a natural task for quantum computers. One prepares the ground state, drives it adiabatically around the twist loop, and reads the accumulated phase off an ancilla [Murta et al., 2020]. The dynamical phase can be cancelled by running the second half of the loop backward in time [Murta et al., 2020]. The obstacle is depth: at a Trotter step that matches the noiseless accuracy of a variational loop, a Trotterized loop for a four-site chain needs $\sim 10^5$–10^6 CNOTs. Mootz and Yao [Mootz and Yao, 2025a] replaced the Trotterized loop by adaptive variational circuits, prepared with AVQITE [Gomes et al., 2021] and propagated with AVQDS [Yao et al., 2021], and recovered the quantized phases of the dimerized Fermi–Hubbard (SSHH) chain with tens to hundreds of layers, about 10^3 times fewer CNOTs. All of those results are noiseless.

Whether that compression buys noise tolerance is not obvious, for two reasons. First, the variational circuit does not carry the global phase, so the Berry phase is assembled from three separately measured pieces: the phase φ_{qc} of a Hadamard-test overlap, and two classically integrated legs,

Preprint.

$\varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle \, dt$ and $\varphi_{G,2} = \int \sum_\mu \dot{\theta}_\mu \langle \hat{A}_\mu \rangle \, dt$, which require thousands of additional expectation values along the loop. Second, the noiseless robustness of the method rests on two cancellations: time reversal removes the dynamical phase, and non-adiabatic errors of $\varphi_{G,2}$ are returned with opposite sign by $\varphi_{\rm qc}$. Both are exact identities for exact data. Nothing guarantees that they survive when the data are noisy, and finite sampling of the readout is irreducible even on a fault-tolerant machine.

We answer this question quantitatively. We re-implement the full pipeline on the SSHH test bed, reproduce the noiseless results and cross-check them against the authors' archived traces, and then execute *every* circuit the method needs (the readout and the 10^5–10^6 circuits per loop that measure the integrands of $\varphi_{G,1}$ and $\varphi_{G,2}$ at $\approx 10^4$ points of the trajectory) in a gate-level density-matrix simulation with depolarizing, relaxation, dephasing and readout errors scaled from a trapped-ion-class profile, and with genuine finite sampling. With a tolerance, estimator and threshold rule fixed in advance, we find:

- **Thresholds.** At 10^4 shots per circuit the quantized phase stays within 0.1π only below a two-qubit error of $p_2^* \approx 9 \times 10^{-5}$ ($U = 0$) and 8×10^{-6} ($U = 10$), 12 and 131 times below today's best two-qubit errors. The interacting circuits are 3.0–5.8$\times$ deeper, but their threshold is 11$\times$ lower.
- **Bias, not shot noise.** The failure is a deterministic drift linear in the noise strength. Shot noise is benign: fewer than ten ($U = 0$) to about 23 ($U = 10$) shots per circuit suffice without gate noise, and more shots do not move the threshold.
- **Mechanism.** The classical bookkeeping, not the depth of any single circuit, is the weak link. Noise pushes the measured energy toward its infinite-temperature value; time reversal cancels about 90% of the resulting dynamical-phase drift, but the residual dominates the error, because noise acts on circuit representations that adaptive growth makes asymmetric between the two halves of the loop. The readout phase is protected by its invariance under contrast loss, whereas the geometric leg contracts generator by generator, so the noiseless $\varphi_{\rm qc}$–$\varphi_{G,2}$ cancellation breaks by an amount set by the gauge-dependent global phase and by how deep in the circuit it is generated. Relaxation during the long circuits matters as much as two-qubit gate errors. Controls confirm the picture: a three times deeper loop drifts no more, because its two halves are more alike, a slower loop is less tolerant, and the same generators grown in a different order change the error by a factor of 1.9.
- **Remedies and baselines.** Zero-noise extrapolation removes most of the bias and raises the threshold by more than a factor of 6.5; a zero-cost depth-matched padding of the time-reversed loop restores the dynamical-phase cancellation at $U = 0$ but not at $U = 10$. Against Trotterized loops under identical noise, the variational method beats the matched-accuracy loop of the original resource comparison (6.0–47$\times$ higher threshold). The quantized phase, however, tolerates a hundred times coarser Trotter step, and that single-circuit loop with $\approx 7 \times 10^3$ CNOTs tolerates 1.5$\times$ ($U = 0$) and 15$\times$ ($U = 10$) more noise than the unmitigated variational method. Because the variational errors are deterministic, ZNE restores the variational method's lead. Learning the trajectory itself from shots is infeasible at any realistic shot count with the method's regularization.

Section 2 summarizes the protocol and our implementation, Sec. 3 specifies the noise model and the pre-registered criteria, and Secs. 4–6 present the anchor, the thresholds and the mechanism. Section 7 treats mitigation, Sec. 8 the comparison with Trotterized loops, Sec. 9 the noisy construction of the trajectory, and Sec. 10 failure modes and costs. Related work is discussed in Sec. 11.

2 The adaptive variational Berry-phase protocol

Test bed. We study the dimerized Fermi–Hubbard (SSHH) ring of Ref. [Mootz and Yao, 2025a],

$$\hat{\mathcal{H}} = \sum_{j,\sigma} \left[t_{j,j+1} \hat{c}^\dagger_{j+1,\sigma} \hat{c}_{j,\sigma} + \text{h.c.} \right] + U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} - \frac{U}{2} \sum_{j\sigma} \hat{n}_{j\sigma}, \qquad t_{j,j+1} = t\,[1 + (-1)^j \delta], \quad (1)$$

with $N = 4$ sites, $t = 1$, periodic boundary conditions and the spin-selective twist $\hat{c}_{j+N,\uparrow} = e^{i\lambda} \hat{c}_{j,\uparrow}$. Our own Jordan–Wigner (JW) map (mode $2j + \sigma$, spin-up on odd qubits) gives $\hat{\mathcal{H}}(\lambda) = \hat{H}_0 + \cos\lambda\, \hat{H}_c + \sin\lambda\, \hat{H}_s$ on 8 qubits with $N_H = 18$ ($U = 0$) or 22 ($U = 10$) non-identity Pauli

strings (12 of weight 3 and 6 of weight 7, plus 4 ZZ terms at $U = 10$), matching the counts of Ref. [Mootz and Yao, 2025a]. We use four configurations at loop time $T = 20$, the shortest loop of the original benchmarks: $U = 0$ with $\delta = \mp 0.3$ and $U = 10$ with $\delta = \mp 0.6$ (the original paper's showcase dimerizations and their mirror images). Exact diagonalization (ED) in the half-filled $(N_\uparrow, N_\downarrow) = (2, 2)$ sector, recomputed at run time, gives a gapped loop (minimum gap 1.20 and 0.883) and the gauge-invariant Wilson-loop Berry phase $\varphi_B^{\mathrm{ED}} = -\arg \prod_k \langle G(\lambda_k)|G(\lambda_{k+1})\rangle$ (2000 points), which is π for $\delta < 0$ (topologically nontrivial) and 0 for $\delta > 0$ (trivial) at both U.

Time-reversed loop and the three-leg estimator. A loop clock $\tau \in [0, T]$ drives $\lambda = 2\pi\tau/T$ while the signed time t runs forward, $t = \tau$, for $\tau \le T/2$ and backward, $t = T - \tau$, afterwards. For a time-reversal-invariant $\hat{\mathcal{H}}$ the dynamical phases of the two halves cancel and the Berry phases add [Murta et al., 2020]. The variational state $|\Psi[\boldsymbol{\theta}]\rangle = \prod_{\mu=1}^{N_\theta} e^{-i\theta_\mu \hat{A}_\mu}|\Psi_0\rangle$ (later generators act later) does not carry the global phase, so the Berry phase is assembled from three separately measured pieces,

$$\varphi_B = \varphi_{\mathrm{qc}} + \varphi_{G,1} + \varphi_{G,2}, \qquad \varphi_{\mathrm{qc}} = \arg\langle \Psi[\boldsymbol{\theta}(0)]|\Psi[\boldsymbol{\theta}(T)]\rangle, \tag{2}$$

$$\varphi_{G,1} = -\int_0^T d\tau\, \frac{dt}{d\tau}\, \langle \hat{\mathcal{H}}(\lambda)\rangle_{\boldsymbol{\theta}}, \qquad \varphi_{G,2} = \int_0^T d\tau \sum_\mu \frac{d\theta_\mu}{d\tau}\, \langle \hat{A}_\mu\rangle_\mu, \tag{3}$$

where $\langle \hat{A}_\mu\rangle_\mu$ is the expectation value of the Pauli generator $\hat{A}_\mu$ in the *prefix* state $\prod_{\nu\le\mu} e^{-i\theta_\nu \hat{A}_\nu}|\Psi_0\rangle$. Equation (3) is the global-phase equation of motion of Ref. [Mootz and Yao, 2025a] written with the identity $\langle\Psi|\partial_{\theta_\mu}\Psi\rangle = -i\langle \hat{A}_\mu\rangle_\mu$; the overall sign convention is fixed by requiring $\varphi_{\mathrm{qc}} + \varphi_G$ to reproduce the phase of the exactly propagated state, which we verified numerically. Two features matter for what follows. (i) Every geometric-connection overlap is a single-Pauli expectation value of a truncated circuit, so it is measured by direct sampling, not by a Hadamard test. (ii) φ_{qc} is a *phase* (a ratio of two quadratures), whereas $\varphi_{G,1}$ and $\varphi_{G,2}$ are *linear functionals of expectation values*: under noise they respond differently (Sec. 6).

State preparation and dynamics (held fixed). The ground state at $\lambda = 0$ is prepared with AVQITE [Gomes et al., 2021] using the 616-element UCCSD-like qubit pool (two- and four-qubit X/Y strings with an odd number of Y [Tang et al., 2021]). The loop is propagated with AVQDS [Yao et al., 2021]: the McLachlan equation $\mathsf{M}\dot{\boldsymbol{\theta}} = \mathbf{V}$ [McLachlan, 1964, Yuan et al., 2019],

$$\mathsf{M}_{\mu\nu} = 2\,\mathrm{Re}\big[\langle\partial_\mu\Psi|\partial_\nu\Psi\rangle - \langle\partial_\mu\Psi|\Psi\rangle\langle\Psi|\partial_\nu\Psi\rangle\big], \quad V_\mu = 2\,\frac{dt}{d\tau}\,\mathrm{Im}\big[\langle\partial_\mu\Psi|\hat{\mathcal{H}}|\Psi\rangle - \langle\partial_\mu\Psi|\Psi\rangle\langle\hat{\mathcal{H}}\rangle\big], \tag{4}$$

is solved with the Tikhonov regularization $\mathsf{M} \to \mathsf{M} + \xi\mathbb{1}$, $\xi = 10^{-6}$, and McLachlan distance $L^2 = 2\,\mathrm{var}\,\hat{\mathcal{H}} - \mathbf{V}^{\mathsf{T}}(\mathsf{M} + \xi)^{-1}\mathbf{V}$ of the original AVQDS work [Yao et al., 2021]. The parameters (including the AVQITE ones) are integrated with RK4 at the adaptive step $\max_\mu |\dot{\theta}_\mu\, \delta\tau| \le \delta\theta_{\max} = 0.01$ (capped at $T/2000$, the archive's output grid). Whenever $L^2 \ge L^2_{\mathrm{cut}} = 10^{-4}$, the generators of the Hamiltonian pool are ranked by the McLachlan distance they would leave, and up to five operators with mutually disjoint support are appended with zero angle (TETRIS growth [Anastasiou et al., 2022, Zhang et al., 2025]), until $L^2 < L^2_{\mathrm{cut}}$. The global phase legs are integrated with the same RK4 stages. All growth decisions use only $(\mathsf{M}, \mathbf{V}, \mathrm{var}\,\hat{\mathcal{H}})$; ED fidelities are computed for diagnostics only.

Implementation choices not fixed by the original paper. Three choices are ours and are documented with their evidence. First, AVQITE uses TETRIS growth with cutoff 10^{-2} (the authors' driver flag); with it our AVQITE yields 20 CNOTs at $U = 0$ (exact ground state) and 92 CNOTs with ground-state infidelity 4.6×10^{-4} at $U = 10$, $\delta = -0.6$, matching the quoted 20 CNOTs, 92 CNOTs and 4.7×10^{-4} (for $\delta = +0.6$: 76 CNOTs, infidelity 5.3×10^{-4}). Second, the reference product state is chosen by a device-measurable rule: AVQITE is run from the prescribed state ($|01011010\rangle$, spin-up on sites 0,1) and from $|11000011\rangle$, and the lower converged energy wins (ties within 0.01 go to the cheaper circuit). From the "wrong" reference AVQITE converges to an excited state (energy error 2.40 at $U = 0$ and 1.83 at $U = 10$), and the rule reproduces the authors' switch of reference for $\delta > 0$. Third, we use the Tikhonov convention of the original AVQDS paper (consistent with the `delta = -1e-06` entry of the archive's run logs). All runs are single-threaded and deterministic.

3 Noise model, estimators, and the threshold criterion

Everything in this section, including the tolerance and the threshold rule, was fixed in a time-stamped analysis plan (`logs/analysis_plan.json`) before any noisy simulation was run. Mitigation schemes (Sec. 7) were designed after the unmitigated results were known and are reported separately; they do not alter the unmitigated thresholds.

Scope: what is noisy. The trajectory $\boldsymbol{\theta}(\tau)$, the generator sequence, and the time grid are frozen at their noiseless values (Sec. 4); the McLachlan integration is not re-run under noise (a separate study in Sec. 9 relaxes this). *Every circuit the method must execute to obtain φ_B is simulated under noise*: the controlled-difference Hadamard test that yields φ_{qc}, and, at all four RK4 stages of every integration step, the circuits that measure $\langle\hat{\mathcal{H}}\rangle$ (for $\varphi_{G,1}$) and the prefix circuits that measure each $\langle\hat{A}_\mu\rangle_\mu$ (for $\varphi_{G,2}$). The integrands are therefore evaluated on exactly the stage grid of the noiseless RK4 integration, with the frozen $d\boldsymbol{\theta}/d\tau$ multiplying the noisy overlaps; in the zero-noise, infinite-shot limit the estimator reproduces the noiseless $\varphi_{G,1}$ and $\varphi_{G,2}$ to 10^{-13}. This stage grid is not optional: the $\varphi_{G,2}$ integrand jumps at every operator insertion (new generators re-partition $d\boldsymbol{\theta}/d\tau$) and at the turning point, and a trapezoid rule on the step nodes alone misses the RK4 value by up to 0.058 rad, while coarser λ grids err by up to 4.08 rad (`logs/quadrature_check.json`). Per configuration this amounts to 8,472–10,276 noisy ansatz states (one per RK4 stage), each read out with one prefix circuit per generator and 10–11 Hamiltonian settings, i.e. 2.5×10^5–1.7×10^6 distinct circuit executions per loop, plus one 9-qubit Hadamard-test circuit pair.

Compilation and schedule. Each factor $e^{-i\theta\hat{P}}$ is compiled as single-qubit basis changes (H for X, $R_X(\pi/2)$ for Y), a CNOT ladder over the support, a virtual (error-free, zero-duration) $R_Z(2\theta)$, the reverse ladder and the inverse basis change, i.e. $2(w-1)$ CNOTs for weight w, which is the all-to-all count of Ref. [Mootz and Yao, 2025a]. Consecutive single-qubit gates on a qubit are merged into one physical gate. In the Hadamard test only the central rotation is controlled by the ancilla, $R_Z(\theta_a{+}\theta_b)\,\mathrm{CX}_{a,q}\,R_Z(\theta_a{-}\theta_b)\,\mathrm{CX}_{a,q}$, so the ancilla selects $\boldsymbol{\theta}(0)$ or $\boldsymbol{\theta}(T)$ at the cost of two CNOTs per parameter (148–1010 CNOTs in total). The overlap $\langle\Psi[\boldsymbol{\theta}(0)]|\Psi[\boldsymbol{\theta}(T)]\rangle$ is read from the ancilla in the X and Y bases. For a prefix measurement of $\hat{A}_\mu$ the final basis change of gadget μ cancels against the measurement basis change, so the prefix circuit ends with the reverse ladder and a Z-basis readout of the support of $\hat{A}_\mu$. Gates are scheduled as soon as possible (all-to-all connectivity; gates on disjoint qubits run in parallel). The ancilla takes part in every controlled rotation, which serializes the Hadamard-test circuit (duration 22–136 ms at the base gate times versus 87 ms for the deepest uncontrolled ansatz).

Noise channels and the device profile. The density matrix of the 8-qubit register (9 with the ancilla) is propagated gate by gate with (i) a two-qubit depolarizing channel of probability p_2 after every CNOT; (ii) a single-qubit depolarizing channel p_1 after every physical single-qubit gate; (iii) thermal relaxation (amplitude damping with T_1 and total dephasing with $T_2 \le 2T_1$) of every qubit over the full wall-clock time it spends in the circuit, busy or idle, with gate durations t_{1q} and t_{2q} (idle relaxation is applied lazily just before the next operation on that qubit, which is exact); and (iv) asymmetric classical readout errors $p(1|0)=e_0$, $p(0|1)=e_1$ on every measured qubit, ancilla and system alike. Because the original resource counts assume all-to-all connectivity, we take as base profile ($s=1$) a trapped-ion-class device: $p_2=10^{-3}$ (average gate infidelity 8×10^{-4}), $p_1=2.5\times10^{-5}$, $e_0=10^{-3}$, $e_1=3\times10^{-3}$, $T_1=10\,\mathrm{s}$, $T_2=0.5\,\mathrm{s}$, $t_{1q}=10\,\mu\mathrm{s}$, $t_{2q}=200\,\mu\mathrm{s}$; these are representative of published Quantinuum H-series component errors [Moses et al., 2023, Quantinuum, 2025], and T_2 is set so that the idle dephasing per two-qubit-gate time (2×10^{-4}) matches the reported per-layer memory error. A single scalar s scales all incoherent rates, $p_{1,2}\to sp_{1,2}$, $e_{0,1}\to se_{0,1}$, $T_{1,2}\to T_{1,2}/s$, at fixed gate times; $s=0$ leaves only finite sampling. The four channels therefore co-vary in the ratio fixed by the profile, and every threshold below is quoted both as s^* and in device units. Channel-resolved runs (Sec. 6) switch on one channel at a time. We sweep s on a logarithmic grid with four points per decade, $10^{-3}\le s\le10^{-0.25}$ plus $s=1$ for $U=10$ and $10^{-2}\le s\le10^{0.25}$ for $U=0$, plus $s=0$ in both cases (the levels $s=10^{-1.25}$ and $10^{-0.75}$–$10^{-0.25}$ at $U=10$ and $s=10^{0.25}$ at $U=0$ were added after the first sweep to evaluate three-point extrapolation; they lie above the unmitigated thresholds and do not affect them).

Estimators and shot model. Every circuit is sampled with S shots (per measurement setting): bit strings are drawn from the exact noisy outcome distribution (multinomial over the measured bits, after readout error), so all shot noise comes from genuine finite sampling. (a) φ_{qc}: two circuits measure the ancilla in the X and Y bases, and $\hat{\varphi}_{\mathrm{qc}} = \mathrm{atan2}(\widehat{\langle Y\rangle}, \widehat{\langle X\rangle})$. This two-quadrature estimator has the same sensitivity at every phase; the single-quadrature readout $P_0 = \frac{1}{2}(1 + \cos\varphi_{\mathrm{qc}})$ of Ref. [Mootz and Yao, 2025a] is evaluated alongside it (with the sign branch supplied by the noiseless value, i.e. in its most favorable form) for comparison. (b) $\langle \hat{A}_\mu \rangle_\mu$: the parity of the Z-readout of the support of $\hat{A}_\mu$ in prefix circuit μ. (c) $\langle \hat{\mathcal{H}} \rangle$: the Pauli strings are grouped into qubit-wise-commuting measurement settings (10 for $U = 0$, 11 for $U = 10$), each sampled with S shots. $\hat{\varphi}_{G,1}$ and $\hat{\varphi}_{G,2}$ follow from Eq. (3) with the RK4 weights of the frozen integration. Seeds are independent sampling realizations (32 per setting); because the density-matrix evolution is deterministic, the exact (infinite-shot) noisy value of every leg is also recorded, which separates deterministic bias from sampling fluctuation. Readout-error mitigation (REM), when reported, inverts the calibrated single-qubit confusion matrices on the same shots.

Accuracy criterion and threshold rule. *Branch policy:* no unwrapping is attempted. The raw sum (2) may lie on any branch (e.g. -5π for $U = 10$); every deviation is computed on the circle, $d(a, b) = |\mathrm{wrap}(a - b)|$ with wrap to $(-\pi, \pi]$, so estimates straddling $\pm\pi$ are compared correctly. The tolerance is $\varepsilon = 0.1\pi$ (0.314 rad, 10% of the separation of the two quantized values). At each (s, S) the accuracy statistic is the root-mean-square circular deviation over seeds, $\mathrm{RMSE} = \langle d(\hat{\varphi}_B, \varphi_B^{\mathrm{ED}})^2\rangle^{1/2}$. The threshold s^* is the first upward crossing of $\mathrm{RMSE}(s) = \varepsilon$ on the ascending grid, interpolated linearly in $\log s$ between the bracketing grid points; its uncertainty is the 16–84% interval of s^* over 2000 bootstrap resamples of the seeds. The reference shot count is $S_{\mathrm{ref}} = 10^4$; device units are $p_2^* = 10^{-3}s^*$, $(e_0, e_1)^* = (10^{-3}, 3{\times}10^{-3})s^*$, $T_1^* = 10\,\mathrm{s}/s^*$, $T_2^* = 0.5\,\mathrm{s}/s^*$. The per-$U$ threshold is the smaller of the trivial and nontrivial values. The shot floor S^* is the smallest S on the grid $\{10, 30, 100, \ldots, 10^6\}$ with $\mathrm{RMSE} \le \varepsilon$ (log-interpolated) at fixed low noise, $s = 0$ (primary) and $s = 0.01$. A single run classifies the topological phase correctly if $d(\hat{\varphi}_B, \varphi_B^{\mathrm{ED}}) < \pi/2$; the two phases of a given U are called distinguishable at (s, S) if at least 95% of the seeds are correct in *both* regimes.

Simulation and validation. All simulators are our own NumPy code. Unit tests confirm that the compiled circuits reproduce the exact ansatz states and prefix expectation values to 10^{-10}, that the controlled-difference Hadamard test reproduces the exact overlap, and that the channels agree with explicit Kraus/Pauli-sum forms (`code/test_dmsim.py`).

4 Noiseless anchor

Before attaching noise we verified that our implementation reproduces the noiseless method (Table 1; `logs/noiseless_*.json`, `logs/crosscheck_archive.json`). All four configurations return the quantized ED value, with circular deviations between 2.1×10^{-5} and 1.1×10^{-4} rad; the $U = 10$ nontrivial loop lands on the same raw branch as the authors' archive ($\varphi_B = -5\pi$ before folding), and the $U = 10$ trivial loop on -6π, illustrating why deviations must be taken on the circle. Gauge-independent observables agree closely with the authors' traces at the same (δ, T): the maximal infidelity to the instantaneous ground state along the loop agrees to within 4.1×10^{-4} (e.g. 0.3345 versus 0.3341 for $U = 10$, $\delta = -0.6$), and the entire $\varphi_{G,1}(\lambda)$ trace agrees to within 2.8×10^{-3} rad once the archive's energy offset $E_G(0)$, which integrates to zero over the loop, is accounted for. Gauge-dependent quantities (φ_{qc}, $\varphi_{G,2}$, and the branch of the raw sum) differ, as they must: they record the adaptive history, which is itself fragile. Perturbing the initial angles by 10^{-13} changes the grown ansatz and $\varphi_{G,2}$ substantially while leaving the folded phase exact (Sec. 6, `logs/variability.json`).

Resources follow the paper's pattern. The peak ansatz has 108/240 CNOTs and depth 12/23 at $U = 0$ (trivial/nontrivial) and 694/670 CNOTs with depth 70/69 at $U = 10$: the interacting circuits are 3.0–5.8× deeper, and at $U = 0$ the nontrivial loop costs 2.2× the CNOTs of the trivial one, whereas at $U = 10$, $\delta = \pm 0.6$ the two regimes cost about the same, as in the archive (944 and 780 CNOTs). Our counts are 0.5–0.9× those of the archive at $L_{\mathrm{cut}}^2 = 10^{-4}$. The archive's own run log records $L_{\mathrm{cut}}^2 = 10^{-5}$ for the $U = 0$, $T = 20$ runs; at that (allowed, tighter) cutoff our nontrivial $U = 0$ loop grows to 784 CNOTs and depth 68, bracketing the archive's 476 CNOTs and depth 47. We keep the

Table 1: Noiseless anchor at $T = 20$, $L_{\text{cut}}^2 = 10^{-4}$ (ours) and the authors' archived traces at the same (δ, T). φ_B: raw sum of Eq. (2) in units of π; d: circular deviation from ED; $1 - f$: maximal infidelity to the instantaneous ED ground state along the loop; N_{CX}, depth: peak ansatz CNOT count and depth (layers of Pauli rotations). Archive CNOT counts exclude the double-counted AVQITE part (Appendix A).

	φ_B (raw)/π		d [rad]		$\max_\lambda(1-f)$		N_{CX} / depth	
Configuration	ours	archive	ours	archive	ours	archive	ours	archive
U=0, δ=+0.3	−0.00003	+2.00002	1.1×10^{-4}	7.6×10^{-5}	0.0232	0.0233	108/12	196/22
U=0, δ=−0.3	−1.00002	+0.99999	7.3×10^{-5}	3.3×10^{-5}	0.0601	0.0603	240/23	476/47
U=10, δ=+0.6	−6.00001	−2.00076	2.1×10^{-5}	2.4×10^{-3}	6.7×10^{-4}	6.5×10^{-4}	694/70	944/104
U=10, δ=−0.6	−4.99998	−5.00016	5.0×10^{-5}	5.0×10^{-4}	0.3345	0.3341	670/69	780/87

paper's stated default $L_{\text{cut}}^2 = 10^{-4}$ for the main study and use the 10^{-5} runs as a controlled depth variation in Sec. 6.

5 Noise thresholds

Thresholds. Figure 1 shows the RMS circular deviation at $S_{\text{ref}} = 10^4$ for all swept noise levels, with the 16–84% band of the per-seed deviations; Table 2 collects the thresholds and representative levels (all levels and seeds are tabulated in Appendix B and logged in `logs/noisy/`). The deviation grows monotonically and essentially linearly with s until it crosses ε; beyond the crossing it eventually wraps around the circle, which is why the error at large noise is not monotonic (e.g. $U = 10$ trivial between $s = 0.316$ and $s = 1$) and why the threshold is defined by the *first* crossing. At $U = 0$ the limiting configuration is the nontrivial one, $s^* = 0.0859$ ($p_2^* = 8.6 \times 10^{-5}$; trivial: $s^* = 0.309$). At $U = 10$ the limiting configuration is the trivial one, $s^* = 0.00762$ ($p_2^* = 7.6 \times 10^{-6}$; nontrivial: $s^* = 0.00974$). In device units, the method stays within 0.1π of the quantized value at 10^4 shots per circuit only below roughly $p_2 \approx 9 \times 10^{-5}$ at $U = 0$ and $p_2 \approx 8 \times 10^{-6}$ at $U = 10$, with readout errors and T_1/T_2 co-scaled as in the profile (Table 2). The bootstrap intervals are narrow because the deviation is dominated by a deterministic bias (next paragraph); the two grid levels that bracket each crossing (a factor $10^{1/4}$ apart) are the rows just below and above the threshold in Table 2. Because the deviation is linear in s, the threshold scales in proportion to the tolerance: halving ε multiplies s^* by 0.48–0.51 and doubling it by 2.01–2.10 (`logs/thresholds.json`), so readers with a different accuracy requirement can rescale our numbers directly. At the base profile itself ($s = 1$, representative of today's best trapped-ion devices), every configuration is far outside the tolerance (RMSE 0.77–2.75 rad).

Bias versus sampling. Below threshold the seed spread of $\hat{\varphi}_B$ at S_{ref} is 0.009–0.015 rad, while the infinite-shot deviation tracks the RMSE to within 0.004 rad: the error is a deterministic bias, not sampling noise. Consistently, raising the shot count does not move the threshold: s^* at $S = 10^3$, 10^4 and 10^5 agrees to within 4% for every configuration (Table 9). Sampling noise alone is benign. With noiseless gates ($s = 0$) the shot floor is $S^* < 10$ at $U = 0$ and $S^* = 18/23$ (trivial/nontrivial) at $U = 10$; at $s = 0.01$ it is $S^* \leq 13$ at $U = 0$, whereas at $U = 10$ that level already sits at (nontrivial) or beyond (trivial) the accuracy threshold. There, more shots cannot help: the trivial loop's RMSE saturates at 0.39 rad even with 10^6 shots, and the nontrivial one hovers at ε (Fig. 2). The floor is low because the global-phase legs average over 10^5–10^6 independently sampled circuits at 8,472–10,276 stages per loop, so the spread of $\hat{\varphi}_B$ is dominated by the single pair of Hadamard-test circuits and falls as $S^{-1/2}$ (Fig. 2c).

Infidelity versus phase error. The output-state infidelity of the deepest ansatz circuit grows linearly with s and reaches 0.048 ($U = 0$) and 0.011 ($U = 10$) at threshold (Fig. 3). The phase error is, to a good approximation, a configuration-dependent multiple of this infidelity: per unit infidelity the $U = 10$ estimate drifts 3.8× more than the $U = 0$ one. At $U = 10$ circuits with an infidelity of only 0.011 already carry a Berry-phase error of 0.1π, so the quantized observable is *more* fragile than the state fidelity suggests. This is the opposite of its noiseless robustness to non-adiabaticity, where loops with infidelities of tens of percent still return the exact phase (Table 1).

Table 2: Main results ($T = 20$, $L^2_{\rm cut} = 10^{-4}$, two-quadrature estimator without mitigation, $S_{\rm ref} = 10^4$ shots per circuit, 32 seeds). Berry phase: circular mean of the folded estimates [rad] (ED: 0 trivial, π nontrivial); d: deviation from ED on the circle (noiseless) or its RMS over seeds [rad]. Threshold: first crossing of RMSE$= \varepsilon = 0.1\pi$, with the 16–84% seed-bootstrap interval, converted to device units of the scaled profile (p_2: two-qubit depolarizing probability; $e_1 = p(0|1)$; T_2; with $e_0 = e_1/3$ and $T_1 = 20\,T_2$). Levels shown: noiseless, shots only, the grid points bracketing the threshold, and the base profile $s = 1$; every level is in Table 7. Depth: layers of Pauli rotations of the deepest ansatz; $N_{\rm CX}$: CNOTs of the deepest ansatz / of the Hadamard test. Noise level s corresponds to $p_2 = 10^{-3}s$.

Configuration	Noise setting	Berry phase	d [rad]	Threshold (device units)	Depth	$N_{\rm CX}$
U=0, triv.	noiseless	-0.00011	1.1×10^{-4}	$s^* = 0.309$ [0.307, 0.311] $p_2^* = 3.1 \times 10^{-4}$, $e_1^* = 9.3 \times 10^{-4}$, $T_2^* = 2$ s	12	108/148
	$s = 0$ (shots)	$+0.002$	0.012			
	$s = 0.178$	$+0.179$	0.180			
	$s = 0.316$	$+0.320$	0.320			
	$s = 1$	$+0.927$	0.927			
U=0, nontriv.	noiseless	$+3.14152$	7.3×10^{-5}	$s^* = 0.0859$ [0.0854, 0.0865] $p_2^* = 8.6 \times 10^{-5}$, $e_1^* = 2.6 \times 10^{-4}$, $T_2^* = 6$ s	23	240/302
	$s = 0$ (shots)	$+3.140$	0.010			
	$s = 0.0562$	-2.936	0.206			
	$s = 0.1$	-2.789	0.353			
	$s = 1$	-0.413	2.729			
U=10, triv.	noiseless	-0.00002	2.1×10^{-5}	$s^* = 0.00762$ [0.00758, 0.00767] $p_2^* = 7.6 \times 10^{-6}$, $e_1^* = 2.3 \times 10^{-5}$, $T_2^* = 66$ s	70	694/1010
	$s = 0$ (shots)	-0.002	0.013			
	$s = 0.00562$	$+0.222$	0.222			
	$s = 0.01$	$+0.396$	0.396			
	$s = 1$	-0.770	0.774			
U=10, nontriv.	noiseless	-3.14154	5.0×10^{-5}	$s^* = 0.00974$ [0.00961, 0.00988] $p_2^* = 9.7 \times 10^{-6}$, $e_1^* = 2.9 \times 10^{-5}$, $T_2^* = 51$ s	69	670/974
	$s = 0$ (shots)	-3.138	0.017			
	$s = 0.00562$	-2.962	0.180			
	$s = 0.01$	-2.822	0.321			
	$s = 1$	-0.390	2.754			

Distinguishing the phases. Classification only requires $d < \pi/2$. Both topological phases are identified correctly by at least 95% of seeds up to $s = 0.325$ at $U = 0$ and $s = 0.0325$ at $U = 10$ (Appendix B), i.e. 3.8× and 4.3× beyond the accuracy threshold; below the accuracy threshold the discrimination margin $\pi - 2d$ exceeds 2.63 rad in every run.

6 Why the threshold sits where it does

Linear response of three legs. Because the error is a deterministic bias, it can be dissected leg by leg with the infinite-shot values that the density-matrix simulation provides (Fig. 4). Every leg responds linearly to the noise scale, $\Delta\varphi_\ell \simeq \beta_\ell s$, up to and beyond the threshold, so each configuration is characterized by three sensitivity coefficients (Table 3). The dynamical leg dominates: $\beta_{G,1}$ accounts for 71–83% of the total bias, the geometric leg $\varphi_{G,2}$ for the rest, while the readout phase contributes at most 1.7%. The two global-phase biases have the same sign, so they add. The linear model predicts the threshold as $s^* \approx \varepsilon/|\beta_{\rm tot}|$, within 4% of the measured value in every configuration (Table 3).

The dynamical-phase cancellation survives only in part. Noise drives $\langle\hat{\mathcal{H}}\rangle$ toward its infinite-temperature value $E_\infty = \mathrm{Tr}\,\hat{\mathcal{H}}/2^8$ (0 at $U = 0$, $-U = -10$ at $U = 10$), raising the measured energy by $\delta E(\tau) \simeq \kappa(\tau)\,[E_\infty - E(\tau)]$ with a contraction κ of the order of the circuit infidelity. Each half of

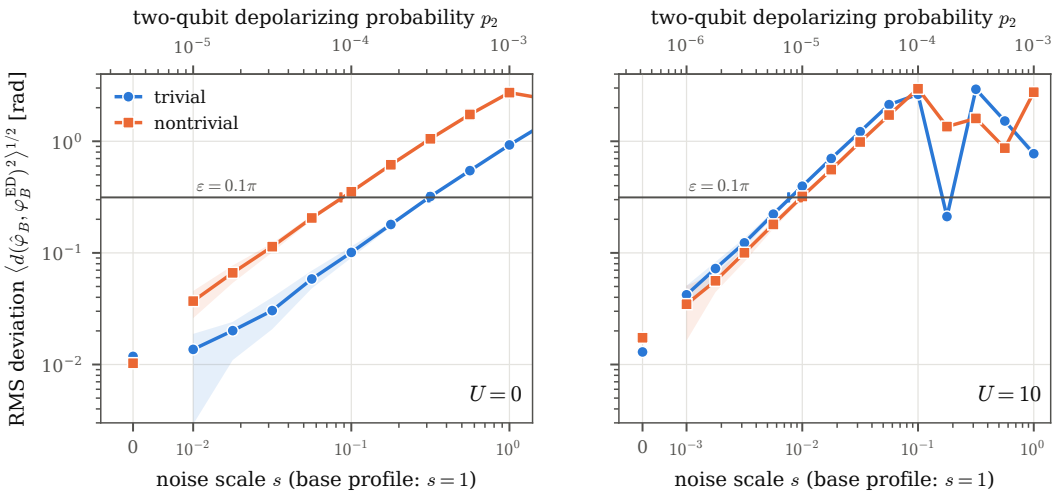


Figure 1: Berry-phase error versus noise strength at $S_{\mathrm{ref}} = 10^4$ shots per circuit. Lines: RMS circular deviation over 32 seeds; bands: 16–84% of the per-seed deviations; leftmost points: $s = 0$ (finite sampling only). Horizontal line: tolerance $\varepsilon = 0.1\pi$; horizontal bars on it: threshold s^* with its bootstrap interval. Top axis: two-qubit depolarizing probability of the scaled profile.

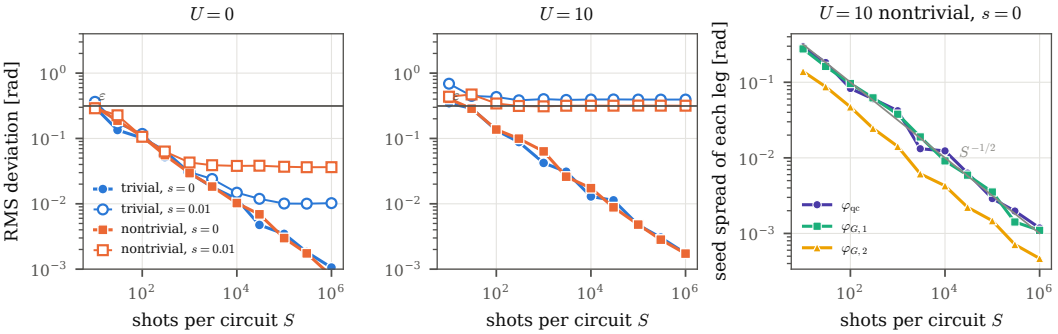


Figure 2: Finite sampling. (a,b) RMS deviation versus shots per circuit S without gate noise ($s = 0$, filled markers) and at $s = 0.01$ (open markers); the horizontal line is ε. (c) Seed spread of each leg for $U = 10$ nontrivial at $s = 0$; the gray line is $S^{-1/2}$. The readout pair dominates the sampling error; the integrated legs average over 10^5–10^6 circuits.

the loop therefore accumulates a large dynamical drift, $\mp \int \delta E \, d\tau \sim \kappa \, |E_\infty - E| \, T/2$: at $s = 0.01$ the two halves drift by 2.5 rad in total for $U = 10$, i.e. several times the tolerance (Fig. 5a, open symbols). Time reversal flips the sign of the second half, and 89–94% of the gross drift cancels. The residual, about a tenth of a quantity that grows with T, $|E_\infty - E|$ and the circuit's error budget, is what crosses ε. Two circuit-level asymmetries produce this residual. First, the ansatz grows during the loop, so the circuit at a backward-half point contains generators that its forward mirror lacks (on average 3.6 extra generators for $U = 0$ nontrivial, 8.8 for $U = 10$ nontrivial). Second, even at equal gate content the backward-half parameters have drifted further from the identity: the rotation weight $\sum_\mu \sin^2 \theta_\mu$ is 19–29% larger at the mirrored point for $U = 0$ and 41–47% larger for $U = 10$ (`logs/rotation_weights.json`). Which of the two matters is decided by how the energy contraction of a circuit depends on its content. A regression of κ over all stage circuits (`logs/kappa_regression.json`) shows that at $U = 0$ the contraction is set by the CNOT count alone ($R^2 = 0.96$–0.998), whereas at $U = 10$ the rotation weight explains far more of it ($R^2 = 0.88$–0.90) than the CNOT count ($R^2 = 0.48$–0.56); with both regressors $R^2 \geq 0.96$ everywhere. Padding every forward circuit with zero-angle copies of the missing generators removes the first asymmetry exactly while leaving the ideal state unchanged. At $U = 0$ this restores the time-reversal cancellation almost completely (net drift $+0.300 \to -0.021$ rad for the nontrivial loop at $s = 0.1$), whereas at $U = 10$ it removes only 16–20% of the net drift, and the forward contraction $\kappa(\tau)$ stays below its

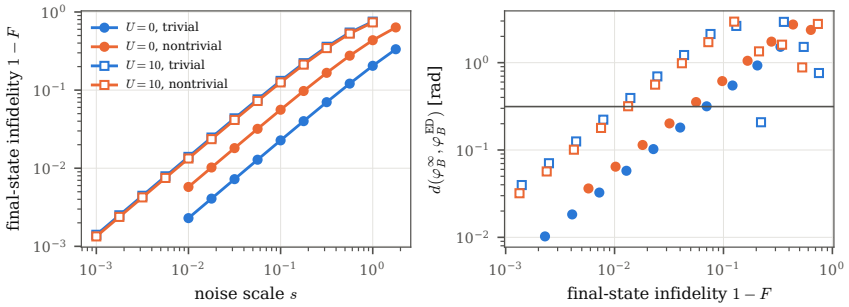


Figure 3: (a) Infidelity of the deepest noisy ansatz circuit (end of the loop) versus noise scale; circles $U = 0$, squares $U = 10$, blue trivial, orange nontrivial. (b) Infinite-shot Berry-phase deviation versus that infidelity for the same runs; the horizontal line is ε.

Table 3: Linear response of the three legs. β_ℓ: least-squares slope [rad per unit s] of the infinite-shot bias $\Delta\varphi_\ell(s)$ over the swept levels $s \le s^*$ (n levels); share of $\varphi_{G,1}$ in $|\beta_{\mathrm{qc}}| + |\beta_{G,1}| + |\beta_{G,2}|$; linear prediction $\varepsilon/|\beta_{\mathrm{tot}}|$ versus the measured threshold.

configuration	β_{qc}	$\beta_{G,1}$	$\beta_{G,2}$	β_{tot}	$G,1$ share	$\varepsilon/\lvert\beta_{\mathrm{tot}}\rvert$	s^*	n
U=0, triv.	−0.018	+0.88	+0.16	+1.02	83%	0.308	0.309	6
U=0, nontriv.	−0.036	+3.05	+0.58	+3.60	83%	0.0874	0.0859	4
U=10, triv.	−0.029	+28.25	+11.36	+39.58	71%	0.00794	0.00762	4
U=10, nontriv.	+0.029	+22.92	+8.88	+31.83	72%	0.00987	0.00974	4

mirror until near the turning point (Fig. 5b). The time-reversal symmetry that cancels the dynamical phase is a symmetry of the physical states; noise acts on their circuit representations, which adaptive growth and parameter drift make asymmetric.

The readout–geometric cancellation breaks. Noiselessly, the gauge-dependent pieces φ_{qc} and $\varphi_{G,2}$ conspire: their sum is the gauge-invariant (Samuel–Bhandari [Samuel and Bhandari, 1988]) geometric phase of the variational path, so any non-adiabatic or reparametrization residual in $\varphi_{G,2}$ is returned with the opposite sign by φ_{qc} (this is the cancellation behind the noiseless robustness of Ref. [Mootz and Yao, 2025a]). Under noise the two legs respond to contraction in opposite ways. φ_{qc} is the argument of an overlap and is invariant under a uniform loss of contrast: its bias stays below 0.089 rad even where the Hadamard-test contrast has fallen to 0.10 ($U = 10$, $s = 1$). $\varphi_{G,2}$ is a linear functional of expectation values, each of which contracts. Writing $\varphi_{G,2} = \sum_\mu C_\mu$ with $C_\mu = \int d\tau\, (d\theta_\mu/d\tau)\langle \hat{A}_\mu\rangle_\mu$, the bias decomposes exactly into per-generator terms $\Delta C_\mu = -\kappa_\mu C_\mu$. We evaluated them on every fourth (at $U = 10$ every eighth) integration step (`logs/prefix_contraction.json`; the sums reproduce the full-grid bias to within 9%). Each contribution contracts with the error of its own prefix circuit: κ_μ rises with the prefix CNOT count (rank correlation 0.81–0.89 in 5 of 6 runs; e.g. $\kappa_\mu = 0.006$–0.015 for the trivial $U = 0$ loops at $s = 0.1$). The bias therefore depends not only on how much global phase the parameters generate, but also on how deep in the circuit it is generated. At $U = 10$ the net phase is dominated by contributions of one sign, and the bias is close to a uniform contraction, $\kappa_{\mathrm{eff}} = -\Delta\varphi_{G,2}/\varphi_{G,2} \approx (0.42\text{–}0.54)\,(1-\bar{F})$ with $\bar{F}$ the loop-averaged circuit fidelity (Fig. 5c); a linear prefix-depth model $\kappa_\mu = a + b\, n_\mu^{\mathrm{CX}}$ predicts it to within 9%. At $U = 0$ large contributions of both signs partly cancel (nontrivial loop: $\varphi_{G,2} = -0.63$ rad but $\sum_\mu |C_\mu| = 12.2$ rad), and the placement of the generators decides the bias. The clearest case is the trivial loop grown at $L_{\mathrm{cut}}^2 = 10^{-5}$, which contains the same generators as the main run in a different order: on the same sampled steps its geometric-leg bias is 3.9× larger and of opposite sign, where a uniform contraction predicts 1.6 (the ratio of the $|\varphi_{G,2}|$) and the prefix-depth model 3.3. For the nontrivial $U = 0$ loops, whose small net phase is a difference of large contributions, neither model predicts the bias reliably (errors up to a factor of 6). In every case the lever arm is the global phase generated by the motion of the variational parameters, $\sum_\mu |C_\mu| = 7.6$–12.2 rad at $U = 0$ and 24.3–26.2 rad at $U = 10$, a gauge-dependent byproduct of the adaptive history rather than a physical property of the loop. The adaptive history itself is fragile. Perturbing the initial angles by 10^{-13} changes the $U = 0$ runs by at most 2.3×10^{-8} rad, but at $U = 10$ it changes the grown ansatz

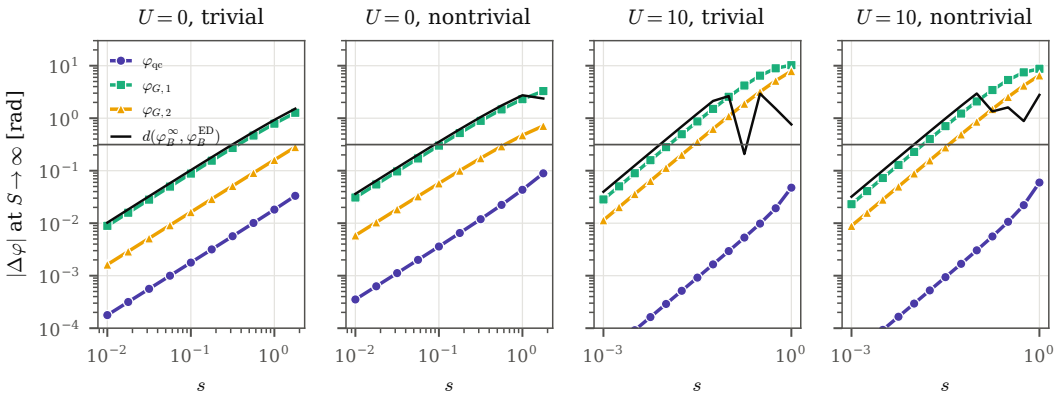


Figure 4: Infinite-shot bias of each leg (absolute value) and resulting circular deviation versus noise scale. All legs respond linearly; the dynamical leg $\varphi_{G,1}$ dominates, the readout phase φ_{qc} is one to two orders of magnitude smaller. The horizontal line is ε.

Table 4: Channel-resolved infinite-shot bias of φ_B [rad] near the threshold, in the linear regime, with all channels and with one channel at a time (rates of the scaled profile); the last column is the sum of the single-channel biases.

configuration	s	all	p_2 only	T_1/T_2 only	p_1 only	readout only	sum
U=0, triv.	0.1	+0.1025	+0.0484	+0.0513	+0.0004	+0.0030	+0.1032
U=0, nontriv.	0.1	+0.3540	+0.1524	+0.1971	+0.0010	+0.0091	+0.3596
U=10, triv.	0.01	+0.3940	+0.1882	+0.2040	+0.0021	+0.0015	+0.3958
U=10, nontriv.	0.01	+0.3170	+0.1524	+0.1630	+0.0018	+0.0010	+0.3183

and its global phase: the trivial loop then grows 678–738 CNOTs with $|\varphi_{G,2}|$ between 10.3 and 15.7 rad (nontrivial loop: 14.8–16.3 rad), and the trivial loop's infinite-shot deviation at $s = 0.01$ ranges over 0.269–0.500 rad (`logs/variability.json`). Because the bias is linear in s, the thresholds of runs that differ only by rounding differ by a factor of 1.9: the noise threshold of an adaptive run is a property of that run's history, not only of the physics.

Deterministic versus stochastic errors. The legs share the frozen trajectory, so their deterministic biases are correlated. Some of this correlated bias cancels (the two halves of $\varphi_{G,1}$, about 90%), and some does not (φ_{qc} versus $\varphi_{G,2}$). A coherent over-rotation of every variational angle, $\theta \to (1+\kappa)\theta$, the cleanest shared deterministic error, behaves the same way (Appendix C): part of it cancels between φ_{qc} and $\varphi_{G,1}$ ($U = 0$ nontrivial at $\kappa = 0.003$: -0.0195 and $+0.0184$ rad), the geometric leg's share scales with $\kappa|\varphi_{G,2}|$, and a miscalibration of only $\kappa^* = 1.0\%$ ($U = 10$) or 2.3% ($U = 0$) exhausts the tolerance. The legs' sampling fluctuations, in contrast, come from disjoint sets of circuits, so they are independent and their variances add: the seed variance of $\hat{\varphi}_B$ agrees with the sum of the leg variances to a median of 8% over all runs, within the $\approx 25\%$ statistical uncertainty of a 32-seed variance. Sharing the trajectory does not make them cancel. The integrated dynamical leg does not random-walk appreciably, because its many stage measurements average down ($\sigma_{G,1} \leq 0.011$ rad at S_{ref}). Its danger is the systematic drift above, which, unlike sampling noise, grows linearly with the loop time.

Which noise source. With one channel at a time in the linear regime (Table 4), two-qubit depolarization produces 48% (43–47%) of the bias at $U = 10$ ($U = 0$), T_1/T_2 relaxation 51–52% (50–56%), and single-qubit errors and readout errors together at most 3%; the channel contributions add to the all-channel bias within 2%. Relaxation and dephasing during the long, partly serialized circuits thus matter as much as the gate errors themselves. Because the contributions add linearly, the threshold can be re-expressed one channel at a time: with ideal memory the method would tolerate $p_2 \lesssim 2.0 \times 10^{-4}$ ($U = 0$) and 1.6×10^{-5} ($U = 10$); with ideal gates it would need $T_2 \gtrsim 3\,\mathrm{s}$ and $34\,\mathrm{s}$ ($T_1 = 20\,T_2$), at the gate times of the profile.

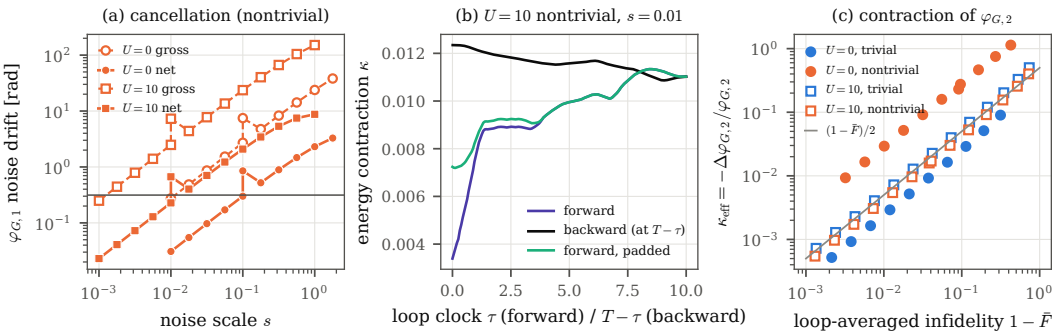


Figure 5: Mechanism. (a) Noise-induced drift of the dynamical leg for the nontrivial configurations (circles $U = 0$, squares $U = 10$): gross drift $|\Delta\varphi_{G,1}^{\text{fwd}}| + |\Delta\varphi_{G,1}^{\text{bwd}}|$ (open, dashed) and net drift after time reversal (filled, solid); about 90% cancels. (b) Energy contraction $\kappa = \delta E/(E_\infty - E)$ of the integrand circuits along the forward half and at the mirrored points of the backward half ($U = 10$ nontrivial, $s = 0.01$, smoothed); padding the forward circuits to the same gate content closes only part of the gap. (c) Effective contraction of the geometric leg versus the loop-averaged circuit infidelity for all swept levels (filled circles $U = 0$, open squares $U = 10$; blue trivial, orange nontrivial); the gray line is a guide.

Which circuit property. Circuit depth, by itself, is not the controlling property. The deeper $U = 10$ circuits (3.0–5.8× the depth) come with a 2.2× larger energy lever arm $\langle|E_\infty - E|\rangle$ and a 5–26× larger $|\varphi_{G,2}|$ (`logs/lever_arms.json`), and it is these lever arms, multiplied by the circuit error, that make the $U = 10$ threshold 11× lower. Three controlled variations of the $U = 0$ loops separate the factors (all at the same physics and noise model). (i) *Deeper circuits.* Grown at $L_{\text{cut}}^2 = 10^{-5}$, the nontrivial loop needs 784 instead of 240 CNOTs and its circuits are 5.6× less faithful, yet its dynamical drift is no larger: at $s = 0.1$ the gross drift is 5.3× larger, but time reversal cancels 98% of it instead of 89%, because its mirrored circuits differ less (the backward-half infidelity exceeds the forward one by 4% instead of 22%), leaving the same net drift (+0.290 versus +0.300 rad). Its geometric-leg bias has the opposite sign (−0.066 versus +0.057 rad), following the sign of its $\varphi_{G,2}$ (+2.88 versus −0.63 rad), and together with the readout bias it offsets the drift, so the deviation stays below ε up to the largest level we ran ($s^* > 0.316$ versus 0.0859). This is compensation, not robustness: at $s = 0.316$, where the circuit infidelity is 46%, the legs contribute +0.66 ($\varphi_{G,1}$), −0.33 ($\varphi_{G,2}$) and −0.13 rad (φ_{qc}). (ii) *Same generators, different history.* The trivial loop grown at $L_{\text{cut}}^2 = 10^{-5}$ selected the same 20 generators (108 CNOTs) in a different order and at different times, and ended with $\varphi_{G,2} = +5.22$ instead of −3.13 rad. Its geometric-leg bias is 3.95× larger and of opposite sign, although $|\varphi_{G,2}|$ differs only by 1.67×, and its threshold is $s^* = 0.629$ versus 0.309. (iii) *Slower loop.* At $T = 100$ the nontrivial loop is shallower (136 CNOTs) and slightly more faithful, but its gross dynamical drift grows with T (4.3× at $s = 0.0562$), the net drift grows from +0.171 to +1.021 rad, and the threshold falls to $s^* = 0.017$: a more adiabatic loop is *less* noise tolerant. The single-ancilla readout, although it is the longest circuit (1010 CNOTs, serialized through the ancilla), is not the bottleneck.

Estimator conditioning. The single-quadrature readout $\varphi_{\text{qc}} = \arccos(2P_0 - 1)$ mistakes any loss of overlap modulus for a phase. This happens even without noise: the non-adiabatic loop leaves $|\langle\Psi(0)|\Psi(T)\rangle| = 0.8166$ at $U = 10$, giving a floor of 0.134 rad at $S = 10^6$ (Fig. 2), and contrast loss adds a bias $\sim \sqrt{2(1-c)}$. It also has vanishing slope where φ_{qc} sits near 0 or π, as it does for the $U = 10$ trivial loop ($\varphi_{\text{qc}} = -3.139$). At S_{ref} its threshold is lower than with two quadratures for the configuration that limits each U (0.0348 vs 0.0859 for $U = 0$ nontrivial, 0.00423 vs 0.00762 for $U = 10$ trivial) and higher for the other two (0.769 and 0.0143), where its contrast-induced bias happens to oppose the global-phase drift. That is an accidental compensation, not robustness. The two-quadrature estimator removes both defects at the cost of one additional circuit.

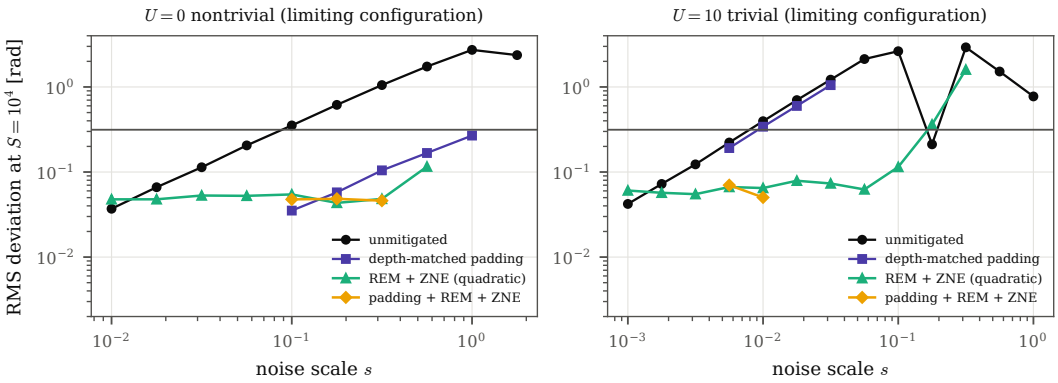


Figure 6: RMS deviation at S_{ref} for the configuration that limits each U: without mitigation, with depth-matched padding, with REM plus quadratic pulse-stretch ZNE, and with padding, REM and ZNE combined (each ZNE point needs the runs at s, $10^{1/4}s$ and $10^{1/2}s$). Horizontal lines: ε.

Table 5: Thresholds s^* at $S_{\mathrm{ref}} = 10^4$ by the same rule with mitigation: readout-error mitigation (REM), depth-matched padding of the forward half (pad), pulse-stretch ZNE after REM with linear (two-point, $\lambda = 1, 10^{1/4}$) or quadratic (three-point, $\lambda = 1, 10^{1/4}, 10^{1/2}$) Richardson extrapolation, and padding combined with quadratic ZNE. "$>$ marks thresholds above the highest level available for that variant, "$<$ below the lowest; –: fewer than two levels at which the variant can be evaluated.

configuration	raw	REM	pad	ZNE lin.	ZNE quad.	pad + ZNE quad.
U=0, triv.	0.309	0.318	> 1	> 1	> 0.562	> 0.316
U=0, nontriv.	0.0859	0.0885	> 1	0.447	> 0.562	> 0.316
U=10, triv.	0.00762	0.00765	0.00902	0.0702	0.159	> 0.01
U=10, nontriv.	0.00974	0.00978	0.011	0.0811	0.183	–

7 Mitigation

The mechanism points to three remedies of increasing generality (Table 5). *Readout-error mitigation* (REM) is free, since it reuses the same shots, but readout is a minor channel and REM changes s^* by at most 3%. *Depth-matched padding* of the forward-half circuits (Sec. 6) costs no extra circuits and removes 89–93% of the net dynamical drift at $U = 0$, raising s^* there by $> 11.6\times$, but only 16–20% at $U = 10$ (gain $1.2\times$). Because the bias is linear in s and has a *deterministic* origin in the contraction of expectation values, *zero-noise extrapolation* attacks it directly. We emulate pulse-stretch ZNE [Kandala et al., 2019] exactly from the noise-scale sweep: the runs at s, λs and $\lambda^2 s$ ($\lambda = 10^{1/4}$) are independent experiments in which every incoherent rate is stretched by λ, and after REM the readout differences drop out. Every measured expectation value (each $\langle \hat{A}_\mu \rangle_\mu$, each Hamiltonian term, and the two Hadamard-test quadratures) is extrapolated by Richardson's rule; since $\varphi_{G,1}$ and $\varphi_{G,2}$ are linear in the measurements, this equals extrapolating the integrated legs. Linear (two-point) extrapolation raises the per-U threshold to $s^*_{\mathrm{ZNE}} = 0.447$ at $U = 0$ and 0.0702 at $U = 10$, i.e. by factors of 5.2 and 9.2. Quadratic (three-point) extrapolation keeps the deviation below ε up to $s^*_{\mathrm{ZNE}} = 0.159$ at $U = 10$ (gain 20.9) and at every level at which the three stretched runs exist at $U = 0$ ($s^*_{\mathrm{ZNE}} > 0.562$, gain > 6.5). The price is variance: below threshold, the extrapolated seed spread is 1.8–$4.8\times$ that of the raw estimate, still far below ε at S_{ref}. Digital ZNE with genuine CNOT folding ($\lambda = 3$), run at one level for each nontrivial configuration (`logs/zne/`), reduces the deviation from 0.316 to 0.011 rad ($U = 10$, $s = 0.01$) and from 0.346 to 0.028 rad ($U = 0$, $s = 0.1$). Folding stretches only the two-qubit gates (their depolarization and the relaxation accumulated while they run), not single-qubit or readout errors, which leaves this small residual.

8 Does compression buy noise tolerance? A Trotter baseline

The practical case for the method is that its shallow circuits should tolerate more noise than a Trotterized adiabatic loop. We test this with first-order Trotterized loops, each executed as a single

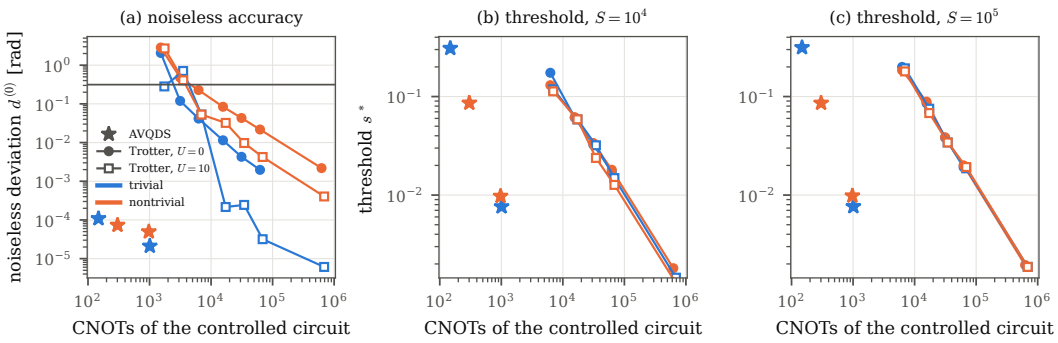


Figure 7: Shallow versus deep. (a) Noiseless circular deviation of first-order Trotterized loops ($\delta t = 2, 1, 0.5, 0.2, 0.1, 0.05, 0.005$ from left to right) versus the CNOTs of the controlled circuit, with the AVQDS loops (stars, at the CNOTs of their Hadamard test) for reference; the horizontal line is ε. (b,c) Threshold s^* at $S = 10^4$ and 10^5 shots per circuit for the Trotter steps that are accurate noiselessly, and for AVQDS (stars). Blue trivial, orange nontrivial; circles $U = 0$, squares $U = 10$.

controlled circuit with the same time-reversed schedule and AVQITE preparation, under the identical noise model, compilation, estimator and threshold rule (Fig. 7, Table 6). The Trotter loop has no classical global-phase legs, and its readout phase enjoys the same contrast invariance as φ_{qc}, so it fails differently: its bias stays close to the noiseless Trotter error while the contrast decays exponentially with the CNOT count, until sampling noise of order $(c\sqrt{S})^{-1}$ swamps the phase.

Matched step. The original resource comparison uses the step $\delta t = 0.005$, whose controlled circuit has 624,020 CNOTs at $U = 0$ ($\approx 2066\times$ the AVQDS Hadamard test) and a noiseless deviation of 0.0022 rad. Against this loop the variational method wins clearly: its per-U threshold is $47\times$ ($U = 0$) and $6.0\times$ ($U = 10$) higher.

Noise-optimal step. The matched step is, however, far finer than the quantized observable requires. The noiseless deviation of the Trotter loop grows with δt (roughly linearly at $U = 0$) but stays within ε up to $\delta t = 0.5$ (6,260–6,972 CNOTs, $d^{(0)} = 0.04$–0.23 rad); only at $\delta t = 1$ do 3 of the four configurations fail (Fig. 7a). A coarser step trades noiseless error for a slower decay of the contrast, and over the accurate range the threshold rises monotonically with δt (Fig. 7b,c; the steps other than 0.005 and 0.05 were added after those results were known, and all use the pre-registered threshold rule). At $\delta t = 0.5$ the Trotter loop reaches $s^* = 0.13$ at $U = 0$ and 0.113 at $U = 10$ (per-U minima), i.e. $1.5\times$ and $15\times$ the AVQDS thresholds; at $S = 10^5$ the factors grow to 2.1 and 24, because the Trotter loop is variance-limited while AVQDS is bias-limited. At $U = 10$ even the step $\delta t = 0.05$, with 68–71$\times$ the CNOTs of the AVQDS Hadamard test, tolerates $1.7\times$ more noise than AVQDS.

Why. The comparison isolates the mechanism of Sec. 6. The variational circuits are shallow, but the estimate integrates 2.5×10^5–1.7×10^6 noisy expectation values per loop, with lever arms (the loop time, the energy distance $|E_\infty - E|$ and the gauge-dependent global phase) that grow with the interaction strength, and the resulting bias does not average away. The Trotter loop reads the phase from a single overlap whose argument is protected against loss of contrast; depth costs it variance, which more shots reduce. Mitigation shifts the balance back. With three-point ZNE, at three times the shots, the variational method's per-U threshold rises to $s^*_{\mathrm{ZNE}} > 0.562$ ($U = 0$) and 0.159 ($U = 10$), $> 4.3\times$ and $1.4\times$ that of the noise-optimal Trotter loop at the same S. Extrapolation is of little use to the Trotter loop, whose errors at threshold are its noiseless Trotter error, which zero-noise extrapolation by construction retains, and the sampling variance of a decayed contrast, which it amplifies. Without mitigation, compression therefore buys noise tolerance only relative to a Trotter step chosen for a noiseless accuracy far beyond what the quantized phase needs; it does not by itself make the Berry-phase estimate more robust. Its advantage lies in the removable, deterministic character of its errors.

Table 6: Shallow versus deep under identical noise, estimator and threshold rule: threshold s^* at $S = 10^4$ shots per circuit ($S = 10^5$ in parentheses). N_{CX}: CNOTs of the controlled (Hadamard-test) circuit at $U = 0$ / $U = 10$ (k $= 10^3$; maximum over the two regimes for the Trotter loops); AVQDS additionally executes 10^5–10^6 shallower integrand circuits per loop. The noiseless deviations of all Trotter steps are within ε (Fig. 7a); $\delta t = 0.005$ is the matched step of the original resource comparison.

method	δt	N_{CX}	U=0, triv.	U=0, nontriv.	U=10, triv.	U=10, nontriv.
AVQDS	–	148–302 / 974–1,010	0.309 (0.315)	0.0859 (0.0859)	0.00762 (0.00765)	0.00974 (0.00982)
Trotter	0.5	6.3k / 7.0k	0.174 (0.200)	0.130 (0.184)	0.118 (0.193)	0.113 (0.180)
Trotter	0.2	15.6k / 17.3k	0.0611 (0.0856)	0.0616 (0.0888)	0.0586 (0.0753)	0.0587 (0.0680)
Trotter	0.1	31.2k / 34.5k	0.0325 (0.0383)	0.0338 (0.0385)	0.0320 (0.0342)	0.0238 (0.0342)
Trotter	0.05	62.4k / 68.9k	0.0173 (0.0192)	0.0181 (0.0200)	0.0149 (0.0187)	0.0127 (0.0193)
Trotter	0.005	624k / 688k	–	0.00181 (0.00195)	0.00145 (0.00187)	0.00128 (0.00186)

9 Learning the trajectory from shots

Everything above freezes the noiseless trajectory. If instead M, $\mathbf{V}$, $\langle\hat{\mathcal{H}}\rangle$ and $\mathrm{var}\,\hat{\mathcal{H}}$ are themselves estimated from S shots per circuit (every elementary expectation value binomially sampled; no gate noise), the McLachlan solve fails long before gate noise matters. Along the frozen trajectories the metric has a median of 4 ($U = 0$) to 33 ($U = 10$) eigenvalues below the Tikhonov parameter $\xi = 10^{-6}$, so sampling noise in those directions is amplified by up to $1/\xi$. The sampled velocity $\dot{\boldsymbol{\theta}}$ is off by a median factor of 7–48 at every $S \leq 10^{12}$, and the sampled McLachlan distance has a spread of 6.5×10^{-4}–2.8×10^{-2} at $S = 10^{12}$ ($L^2_{\mathrm{cut}} = 10^{-4}$), so growth decisions disagree with the noiseless ones in 16–48% of the cases even at 10^{12} shots per circuit (`logs/construction_noise.json`). Larger regularization trades variance for bias. At $\xi = 10^{-2}$ the velocity error falls to 2% at $S = 10^{10}$ for $U = 0$, but the regularization itself changes the noiseless velocity by 23% ($U = 0$) and 98% ($U = 10$) (`logs/construction_xi.json`). Full stochastic integrations confirm the picture (`logs/noisy_construction/`). In all 4 guarded runs of the $U = 0$ trivial loop ($S = 10^8$–10^{12}, $\xi = 10^{-6}$ and 10^{-2}) the sampled McLachlan distance keeps triggering growth: after 4,000 McLachlan evaluations the ansatz had grown to 393–833 generators (20 in the noiseless run) while the loop had advanced only to $\tau = 0.08$–9.3 of $T = 20$. The frozen-trajectory scope of this study is therefore the only one in which the method can be characterized at realistic shot counts; constructing the trajectory on hardware would require a reformulation of the McLachlan step (e.g. stronger, noise-aware regularization or a different metric estimator), which is beyond this work.

10 Failure modes and computational cost

How the estimate fails. We observed six distinct failure modes; each is documented in the logs. (1) *Systematic drift beyond threshold.* The bias grows linearly in s and eventually exceeds $\pi/2$, at which point the topological phase is misclassified with certainty rather than at random, because the seeds agree with one another while being wrong. Beyond that the estimate wraps around the circle and the error stops growing monotonically ($U = 10$ trivial: RMSE 2.92 rad at $s = 0.316$ but 0.77 rad at $s = 1$), so a smaller error at a larger noise level is not evidence of robustness. (2) *Ill-conditioned readout.* The single-quadrature estimator inherits a floor from the non-adiabatic overlap modulus and a $\sqrt{1-c}$ bias from contrast loss (Sec. 6). It also loses the sign of φ_{qc}, which matters whenever $\varphi_G \neq 0$. (3) *Quadrature on a coarse grid.* The $\varphi_{G,2}$ integrand jumps at operator insertions, so measuring it on a coarse λ grid instead of the RK4 stages introduces errors of up to 4.08 rad that do not vanish with more shots. (4) *Branch handling.* The raw sums of the four loops lie on four different branches (Table 1), and for the $U = 10$ trivial loop φ_{qc} sits at $\pm\pi$, so any unwrapping heuristic or deviation taken off the circle would produce spurious 2π errors. (5) *History dependence.* $|\varphi_{G,2}|$ and the depth are outcomes of the adaptive growth; they change between numerically equivalent runs (`logs/variability.json`) and move the threshold with them. (6) *Noisy construction.* When the trajectory itself is learned from shots, growth decisions become random and the integration stalls or over-grows the ansatz (Sec. 9).

Cost of the method under noise. Per loop the method executes, besides the Hadamard-test pair, one prefix circuit per generator and 10–11 Hamiltonian measurement settings at each of the 8,472–10,276

RK4 stages: 2.5×10^5 ($U = 0$ nontrivial) to 1.7×10^6 ($U = 10$ trivial) distinct circuit executions. Because the threshold is bias-limited, the sampling error needs few shots: the cheapest allocations that stay within ε without gate noise use 10 shots per integrand circuit and 10–100 per Hadamard-test circuit, i.e. 2.5×10^6 ($U = 0$) to 1.7×10^7 ($U = 10$) shots per loop (`logs/shot_allocation.json`). A comfortable $S = 10^3$, with a seed spread of 0.029–0.063 rad, costs 10^8–10^9 shots per loop, three times as many with three-point ZNE. All of this assumes a noiselessly known trajectory. Learning the trajectory on the device needs, at every RK4 stage, at least the $N_\theta(N_\theta + 1)/2$ metric entries and the N_θ entries of $\mathbf{V}$, i.e. 1.8×10^6–1.2×10^8 circuits per loop (7–73$\times$ the evaluation circuits), and at the far larger shot counts of Sec. 9. The Trotter baseline has a single circuit per loop but needs $S \propto c^{-2}$, which grows exponentially with the CNOT count.

Cost of this study. All density-matrix simulations were run on CPUs (at most 32 concurrent single-threaded worker processes). One noise level costs 0.4–0.4 CPU-hours at $U = 0$ and 1.9–2.1 CPU-hours at $U = 10$ (8,472–10,276 noisy 8-qubit ansatz states of up to 694 CNOTs plus one 9-qubit Hadamard test of up to 1010 CNOTs). The whole study used 139 CPU-hours (`logs/cost_summary.json`). Finite sampling is cheap by comparison, because the exact noisy outcome distributions are reused for all seeds and shot counts.

11 Related work

Quantum algorithms for Berry phases. Cyclic adiabatic evolution with a time-reversed second half, read out by a Hadamard test, was proposed by Murta *et al.* [Murta et al., 2020]; Mootz and Yao replaced the Trotterized loop by AVQDS and AVQITE circuits and supplied the classical global-phase legs that the variational representation requires [Mootz and Yao, 2025a], demonstrating noiseless accuracy at depths of tens to hundreds of layers. Hayakawa *et al.* recently gave a full-range Berry-phase estimation algorithm without time-reversal symmetry and settled its complexity [Hayakawa et al., 2025]. Variational routes to topological invariants that avoid dynamics altogether include VQE-based Berry phases and nonadiabatic couplings [Tamiya et al., 2021] and the noise-robust overlap-winding construction of Xiao, Freericks and Kemper [Xiao et al., 2023], which was run on IBM hardware; randomized-measurement [Elben et al., 2020] and string-order [Azses et al., 2020, Smith et al., 2022] protocols probe topology from snapshots. None of these studies quantifies how the variational global-phase bookkeeping of the AVQDS protocol behaves under gate noise, which is our subject.

Noise and variational dynamics. The McLachlan principle [McLachlan, 1964, Yuan et al., 2019] underlies AVQDS [Yao et al., 2021] and its TETRIS-compressed variant [Zhang et al., 2025, Anastasiou et al., 2022]. Shot-noise propagation through the metric inversion of variational dynamics was analyzed by van Straaten and Koczor [van Straaten and Koczor, 2021]; Fontana *et al.* characterized the noise resilience of variational eigensolvers and found linear degradation below a circuit-dependent threshold [Fontana et al., 2021]; projected (pVQD) [Barison et al., 2021] and open-system [Endo et al., 2020] variational dynamics address related trade-offs. Generic entropic arguments bound the useful depth of noisy circuits [Stilck França and García-Patrón, 2021]. Our analysis differs in focusing on a quantity assembled from three differently-estimated pieces, and in showing that the dominant failure is a *deterministic* bias generated by the method's classical integration of noisy expectation values rather than shot noise or state infidelity per se.

Geometric phases and phase estimation under noise. Geometric phases of open systems are insensitive to pure dephasing in idealized settings [Carollo et al., 2003] and robust to slow classical fluctuations [De Chiara and Palma, 2003]; equivalence conditions for the noise sensitivity of geometric and dynamical gates were derived in [Colmenar et al., 2022]. Single-ancilla phase estimation is known to retain phase information under depolarizing noise, which only reduces the contrast [O'Brien et al., 2019]; our two-quadrature readout confirms this in a realistic gate-level setting. The Samuel–Bhandari (Pancharatnam) phase [Samuel and Bhandari, 1988] provides the gauge-invariant reading of $\varphi_{\mathrm{qc}} + \varphi_{G,2}$ we use in Sec. 6.

Error mitigation and Trotter baselines. Zero-noise extrapolation [Temme et al., 2017, Li and Benjamin, 2017] by pulse stretching [Kandala et al., 2019] or unitary folding [Giurgica-Tiron et al., 2020], noise-estimation circuits [Urbanek et al., 2021], and readout-error inversion [Bravyi et al.,

2021] are reviewed in [Cai et al., 2023]; we apply readout inversion and digital ZNE and introduce a mitigation specific to time-reversed loops (depth-matched padding). The trade-off between Trotter error and accumulated gate error was analyzed by Knee and Munro [Knee and Munro, 2015]; we measure it directly for the Berry-phase loop against the adaptive variational circuits.

12 Limitations and conclusions

Limitations. (i) *Frozen trajectory.* The thresholds characterize the measured Berry phase along the noiselessly constructed trajectory. Learning that trajectory on hardware is a harder problem, which Sec. 9 shows to be out of reach at realistic shot counts with the method's regularization. (ii) *Noise model.* Our channels are Markovian and local, with all-to-all connectivity and a single co-scaling knob; crosstalk, leakage, non-Markovian drift and connectivity-induced SWAP overhead (which would deepen every circuit on a planar device) are not modeled, and coherent errors are covered only by a systematic over-rotation study (Appendix C). Because the bias is linear in every channel, the thresholds for another device can be estimated from the per-channel sensitivities of Table 4. (iii) *Scope of configurations.* We study one loop time ($T = 20$, with a $T = 100$ control), one cutoff (10^{-4}, with a 10^{-5} control), and the four configurations of the original study; the history dependence of $|\varphi_{G,2}|$ means that individual adaptive runs at other parameters may sit somewhat above or below our thresholds. (iv) *Mitigation.* Pulse-stretch ZNE is emulated under the assumption that every incoherent rate scales exactly with the stretch factor; the digital-folding check stretches only the two-qubit gates. Real devices scale noise imperfectly. (v) *Baseline.* The Trotter comparison uses first-order product formulas at our four operating points with all-to-all connectivity. How coarse a step stays accurate depends on the gap and the loop time, and SWAP overhead on a planar device would add to the long Trotter circuit and to the many short variational circuits in different proportions.

Conclusions. The adaptive variational Berry-phase algorithm compresses the circuits by three orders of magnitude, but under noise its accuracy is governed by the classical integration of noisy expectation values rather than by circuit depth. Its quantized output stays within 0.1π of the exact value only below $p_2 \approx 9 \times 10^{-5}$ for the noninteracting chain and $p_2 \approx 8 \times 10^{-6}$ in the strongly correlated regime, with relaxation during the circuits contributing as much as the gate errors. The limiting error is a deterministic drift that more shots cannot remove. It arises because the cancellations that make the method robust to non-adiabaticity are identities about states, while noise acts on their circuit representations: time reversal still removes about 90% of the dynamical-phase drift, but the remainder, which grows with the loop time, and the generator-by-generator contraction of the geometric leg, which grows with the gauge-dependent variational global phase and with the depth at which it is generated, set the threshold. The readout phase is not the weak link, provided it is estimated from both quadratures. Measured against the Trotterized loop it was designed to replace, the method wins only if that loop uses the fine step chosen for noiseless accuracy; because the quantized phase tolerates a hundred times coarser steps, a single deep Trotter circuit is, without mitigation, the more noise-tolerant way to obtain the same answer at both interaction strengths in our simulations, and decisively so at $U = 10$. Because the variational method's errors are deterministic, zero-noise extrapolation restores its lead (by $> 4.3\times$ at $U = 0$ and $1.4\times$ at $U = 10$, at three times the shots). Practical implications follow directly. One should (a) measure both quadratures, (b) keep loops short and the variational global phase small (for example by gauge-fixing the adaptive ansatz so that $\varphi_{G,2}$ stays near zero), (c) apply zero-noise extrapolation to the global-phase legs, which in our simulations raises the tolerable noise by more than a factor of 6.5, and (d) benchmark resource claims against the coarsest Trotter step that still meets the accuracy target rather than against a step matched to the variational method's noiseless accuracy.

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A Archive inspection and noiseless cross-check

The authors' figshare archive [Mootz and Yao, 2025b] was read directly from the zip file and used only for validation (`codebase_notes.md`). Its `results<run>p0.h5` files store, on a 2000-point λ grid, the AVQDS and ED energies (shifted by $E_G(0)$), the instantaneous ground-state fidelity, the traces of φ_{qc} (`th1`), $\varphi_{G,1}$ (`th7`) and $\varphi_{G,2}$ (`th5`), with `th4` $=$ `th5` $+$ `th7` $=\varphi_G$, and operator counts by Pauli weight. We fold both pipelines in the same way: the raw sum $\varphi_{\mathrm{qc}}+\varphi_G$ at $\lambda=2\pi$ is mapped to $(-\pi,\pi]$ and compared with ED on the circle (the authors' scripts plot $|\mathrm{wrap}(-(\texttt{th4}+\texttt{th1}))|$, which is equivalent for the quantized values). Two bookkeeping details matter when comparing resources. First, the figure scripts add the AVQITE CNOTs to an operator count that already includes them, so, e.g., the quoted 496 CNOTs of the $U=0$, $\delta=-0.3$, $T=20$ run correspond to 476 actual CNOTs. Second, the archived $\varphi_{G,1}$ is integrated with $\hat{\mathcal{H}}-E_G(0)$, which shifts the trace by $E_G(0)\,t$ but not its end point. The run logs record $L^2_{\mathrm{cut}}=10^{-5}$ for the $U=0$, $T=20$ runs and a Tikhonov parameter 10^{-6}. Figure 8 overlays our traces on the archive's.

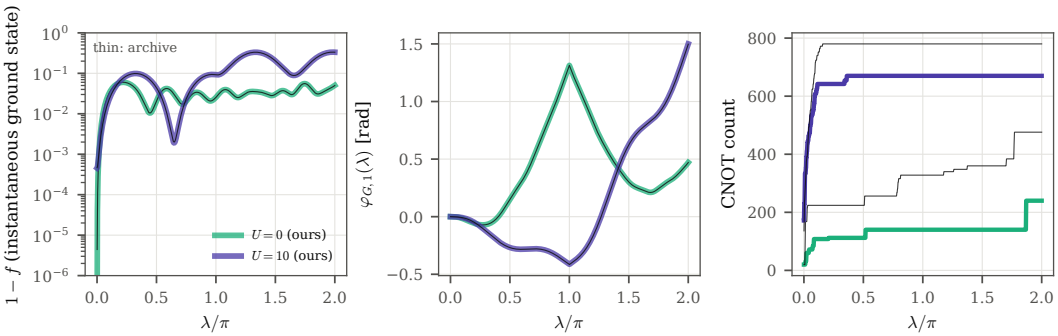


Figure 8: Noiseless anchor versus the authors' archive at the same $(\delta, T=20)$ for the nontrivial configurations. Thick colored lines: ours; thin black lines: archive (they lie on top of ours in the first two panels). Left: infidelity to the instantaneous ED ground state; middle: $\varphi_{G,1}(\lambda)$ (archive offset $E_G(0)t$ removed); right: CNOT count of the growing ansatz (ours at $L^2_{\mathrm{cut}}=10^{-4}$, archive at 10^{-5}).

B All noise levels and seeds

Table 7 lists every swept noise level of the main study at $S_{\mathrm{ref}}=10^4$ (32 seeds each); per-seed estimates for all three shot counts are in `logs/noisy/<config>__s<s>__all__f1__k0.json`. Table 9 gives the threshold at $S=10^3, 10^4, 10^5$, and Table 10 the fraction of seeds that classify both topological phases correctly. Table 8 lists every other noisy whole-method run (padding, controls, history variants, single channels, folding).

C Coherent over-rotation

A systematic multiplicative miscalibration of every variational rotation, $\theta\to(1+\kappa)\theta$, applied consistently to the integrand circuits and to both branches of the Hadamard test (no incoherent noise, $S\to\infty$), produces a deviation that grows approximately linearly for $\kappa\le 0.003$ and faster beyond (Table 11). Part of this shared deterministic error cancels between the legs: for $U=0$ nontrivial at $\kappa=0.003$, φ_{qc} and $\varphi_{G,1}$ shift by -0.0195 and $+0.0184$ rad, leaving a total deviation of 0.0072 rad. The geometric leg does not take part in this cancellation, because the estimator multiplies the over-rotated overlaps by the *frozen* $\dot{\boldsymbol{\theta}}$. Its bias is $(0.56\text{–}1.09)\,\kappa|\varphi_{G,2}|$ at $\kappa=0.003$ in three configurations, the same lever arm as for incoherent noise; the exception is $U=0$ nontrivial, whose small net $\varphi_{G,2}$ is the difference of large contributions of both signs (Sec. 6). For this error alone the accuracy threshold (first crossing of the infinite-shot deviation with ε, log-interpolated in κ, minimum over regimes) is $\kappa^*=2.3\%$ at $U=0$ and 1.0% at $U=10$.

Table 7: All swept noise levels ($S = 10^4$, 32 seeds). RMSE, median and 90th percentile of the circular deviation d [rad]; f_ε: fraction of seeds with $d \le \varepsilon$; f_{cls}: fraction with $d < \pi/2$; d^∞: infinite-shot deviation; $\Delta\varphi_\ell$: infinite-shot bias of each leg; σ: seed spread of $\hat{\varphi}_B$; $1 - F$: final-state infidelity; c: Hadamard-test contrast.

config.	s	RMSE	med.	p90	f_ε	f_{cls}	d^∞	$\Delta\varphi_{\mathrm{qc}}$	$\Delta\varphi_{G,1}$	$\Delta\varphi_{G,2}$	σ	$1-F$	c
U=0, triv.	0	0.012	0.009	0.019	1.00	1.00	0.000	+0.0000	+0.000	+0.000	0.012	0.0000	1.000
U=0, triv.	0.01	0.014	0.010	0.021	1.00	1.00	0.010	−0.0002	+0.009	+0.002	0.012	0.0023	0.997
U=0, triv.	0.0178	0.020	0.017	0.028	1.00	1.00	0.018	−0.0003	+0.016	+0.003	0.010	0.0041	0.995
U=0, triv.	0.0316	0.030	0.027	0.042	1.00	1.00	0.033	−0.0006	+0.028	+0.005	0.010	0.0072	0.991
U=0, triv.	0.0562	0.058	0.056	0.071	1.00	1.00	0.058	−0.0010	+0.050	+0.009	0.011	0.0128	0.985
U=0, triv.	0.1	0.101	0.101	0.115	1.00	1.00	0.102	−0.0018	+0.088	+0.016	0.012	0.0227	0.973
U=0, triv.	0.178	0.180	0.179	0.190	1.00	1.00	0.181	−0.0032	+0.155	+0.029	0.009	0.0400	0.952
U=0, triv.	0.316	0.320	0.319	0.333	0.28	1.00	0.316	−0.0056	+0.271	+0.051	0.010	0.0701	0.916
U=0, triv.	0.562	0.546	0.547	0.559	0.00	1.00	0.548	−0.0101	+0.467	+0.091	0.012	0.1211	0.856
U=0, triv.	1	0.927	0.928	0.943	0.00	1.00	0.929	−0.0182	+0.786	+0.160	0.012	0.2047	0.758
U=0, triv.	1.78	1.520	1.517	1.538	0.00	1.00	1.517	−0.0332	+1.269	+0.282	0.014	0.3336	0.610
U=0, nontriv.	0	0.010	0.008	0.016	1.00	1.00	0.000	+0.0000	+0.000	+0.000	0.010	0.0000	1.000
U=0, nontriv.	0.01	0.037	0.037	0.047	1.00	1.00	0.036	−0.0004	+0.031	+0.006	0.010	0.0058	0.994
U=0, nontriv.	0.0178	0.066	0.064	0.080	1.00	1.00	0.064	−0.0006	+0.055	+0.010	0.012	0.0102	0.989
U=0, nontriv.	0.0316	0.114	0.114	0.124	1.00	1.00	0.114	−0.0011	+0.097	+0.018	0.010	0.0181	0.981
U=0, nontriv.	0.0562	0.206	0.205	0.222	1.00	1.00	0.202	−0.0020	+0.171	+0.033	0.011	0.0319	0.966
U=0, nontriv.	0.1	0.353	0.353	0.366	0.00	1.00	0.354	−0.0036	+0.300	+0.057	0.012	0.0561	0.941
U=0, nontriv.	0.178	0.616	0.613	0.633	0.00	1.00	0.615	−0.0065	+0.522	+0.100	0.011	0.0974	0.898
U=0, nontriv.	0.316	1.052	1.052	1.066	0.00	1.00	1.051	−0.0119	+0.890	+0.173	0.012	0.1664	0.826
U=0, nontriv.	0.562	1.739	1.736	1.752	0.00	0.00	1.741	−0.0223	+1.472	+0.292	0.013	0.2759	0.713
U=0, nontriv.	1	2.729	2.729	2.754	0.00	0.00	2.734	−0.0433	+2.304	+0.473	0.022	0.4351	0.552
U=0, nontriv.	1.78	2.369	2.373	2.405	0.00	0.00	2.371	−0.0891	+3.287	+0.714	0.029	0.6343	0.355
U=10, triv.	0	0.013	0.008	0.022	1.00	1.00	0.000	+0.0000	+0.000	+0.000	0.013	0.0000	1.000
U=10, triv.	0.001	0.042	0.039	0.060	1.00	1.00	0.040	−0.0000	+0.028	+0.011	0.013	0.0014	0.998
U=10, triv.	0.00178	0.072	0.073	0.085	1.00	1.00	0.071	−0.0001	+0.050	+0.020	0.012	0.0025	0.996
U=10, triv.	0.00316	0.123	0.121	0.138	1.00	1.00	0.125	−0.0001	+0.089	+0.036	0.011	0.0044	0.993
U=10, triv.	0.00562	0.222	0.221	0.240	1.00	1.00	0.222	−0.0002	+0.159	+0.064	0.014	0.0079	0.987
U=10, triv.	0.01	0.396	0.399	0.414	0.00	1.00	0.394	−0.0003	+0.281	+0.113	0.016	0.0140	0.977
U=10, triv.	0.0178	0.701	0.699	0.718	0.00	1.00	0.696	−0.0005	+0.496	+0.201	0.013	0.0247	0.960
U=10, triv.	0.0316	1.219	1.217	1.245	0.00	1.00	1.223	−0.0009	+0.868	+0.356	0.019	0.0435	0.930
U=10, triv.	0.0562	2.129	2.131	2.146	0.00	0.00	2.129	−0.0016	+1.505	+0.626	0.016	0.0760	0.879
U=10, triv.	0.1	2.632	2.630	2.652	0.00	0.00	2.635	−0.0029	+2.557	+1.094	0.015	0.1312	0.795
U=10, triv.	0.178	0.212	0.213	0.231	1.00	1.00	0.208	−0.0053	+4.193	+1.888	0.020	0.2211	0.664
U=10, triv.	0.316	2.923	2.928	2.947	0.00	0.00	2.931	−0.0098	+6.461	+3.184	0.027	0.3586	0.483
U=10, triv.	0.562	1.519	1.511	1.559	0.00	0.91	1.515	−0.0193	+8.931	+5.169	0.032	0.5454	0.274
U=10, triv.	1	0.774	0.776	0.872	0.00	1.00	0.761	−0.0475	+10.244	+7.892	0.078	0.7526	0.099
U=10, nontriv.	0	0.017	0.010	0.033	1.00	1.00	0.000	+0.0000	+0.000	+0.000	0.017	0.0000	1.000
U=10, nontriv.	0.001	0.035	0.028	0.053	1.00	1.00	0.032	+0.0000	+0.023	+0.009	0.015	0.0013	0.998
U=10, nontriv.	0.00178	0.056	0.057	0.071	1.00	1.00	0.057	+0.0001	+0.041	+0.016	0.013	0.0024	0.996
U=10, nontriv.	0.00316	0.100	0.100	0.117	1.00	1.00	0.101	+0.0001	+0.073	+0.028	0.012	0.0042	0.993
U=10, nontriv.	0.00562	0.180	0.178	0.196	1.00	1.00	0.179	+0.0002	+0.129	+0.050	0.014	0.0075	0.988
U=10, nontriv.	0.01	0.321	0.317	0.343	0.41	1.00	0.317	+0.0003	+0.228	+0.089	0.021	0.0133	0.978
U=10, nontriv.	0.0178	0.557	0.555	0.575	0.00	1.00	0.560	+0.0005	+0.402	+0.157	0.014	0.0236	0.962
U=10, nontriv.	0.0316	0.985	0.986	1.003	0.00	1.00	0.985	+0.0009	+0.706	+0.278	0.016	0.0415	0.933
U=10, nontriv.	0.0562	1.719	1.716	1.739	0.00	0.00	1.718	+0.0017	+1.225	+0.491	0.018	0.0726	0.884
U=10, nontriv.	0.1	2.955	2.954	2.973	0.00	0.00	2.951	+0.0030	+2.087	+0.860	0.016	0.1254	0.804
U=10, nontriv.	0.178	1.354	1.351	1.379	0.00	1.00	1.347	+0.0056	+3.439	+1.492	0.019	0.2120	0.678
U=10, nontriv.	0.316	1.603	1.605	1.632	0.00	0.16	1.607	+0.0106	+5.342	+2.537	0.027	0.3451	0.501
U=10, nontriv.	0.562	0.868	0.869	0.909	0.00	1.00	0.882	+0.0221	+7.482	+4.180	0.041	0.5282	0.293
U=10, nontriv.	1	2.754	2.760	2.904	0.00	0.00	2.777	+0.0595	+8.745	+6.539	0.107	0.7357	0.113

D Reproducibility

All code is in `code/` (Python 3.12, NumPy, SciPy): `sshh.py` (model and Jordan–Wigner map), `ed_reference.py` (ED references), `variational.py`, `avq.py` (AVQITE/AVQDS), `run_noiseless.py`, `dmsim.py` (compiled-circuit density-matrix simulator), `noisy_eval.py` and `run_sweep.py` (noisy whole-method evaluation), `trotter*.py`, `zne.py`, `noisy_construction.py`, `construction_*.py`, `analyze_*.py`, `mechanism.py`, `prefix_contraction.py`, `make_*.py` (figures, tables and the macro file from which every number in this paper is typeset); the run scripts `seg*.sh` and `lane*.sh` list every sweep, and `postprocess.sh` regenerates all derived logs, tables, figures and macros from the raw run logs. Unit tests: `test_dmsim.py` (`logs/unit_tests.txt`). The analysis plan with the tolerance and threshold rule is `logs/analysis_plan.json`; it predates all noisy runs.

Table 8: Every other noisy whole-method run ($S = 10^4$, 32 seeds; shot-grid runs are shown in Fig. 2, over-rotation runs in Table 11). Columns as in Table 7.

config.	variant	s	RMSE	d^{∞}	$\Delta\varphi_{\mathrm{qc}}$	$\Delta\varphi_{G,1}$	$\Delta\varphi_{G,2}$	$1-F$	c
U=0, triv.	padded	0.1	0.027	0.024	-0.0018	$+0.010$	$+0.016$	0.0227	0.973
U=0, triv.	padded	0.178	0.042	0.043	-0.0032	$+0.017$	$+0.029$	0.0400	0.952
U=0, triv.	padded	0.316	0.074	0.075	-0.0056	$+0.030$	$+0.051$	0.0701	0.916
U=0, triv.	padded	0.562	0.128	0.132	-0.0101	$+0.051$	$+0.091$	0.1211	0.856
U=0, triv.	padded	1	0.229	0.226	-0.0182	$+0.084$	$+0.160$	0.2047	0.758
U=0, nontriv.	padded	0.1	0.035	0.033	-0.0036	-0.021	$+0.057$	0.0561	0.941
U=0, nontriv.	padded	0.178	0.058	0.058	-0.0065	-0.036	$+0.100$	0.0974	0.898
U=0, nontriv.	padded	0.316	0.105	0.100	-0.0119	-0.061	$+0.173$	0.1664	0.826
U=0, nontriv.	padded	0.562	0.167	0.168	-0.0223	-0.102	$+0.292$	0.2759	0.713
U=0, nontriv.	padded	1	0.268	0.269	-0.0433	-0.160	$+0.473$	0.4351	0.552
U=10, triv.	padded	0.00562	0.191	0.190	-0.0002	$+0.127$	$+0.064$	0.0079	0.987
U=10, triv.	padded	0.01	0.341	0.337	-0.0003	$+0.224$	$+0.113$	0.0140	0.977
U=10, triv.	padded	0.0178	0.598	0.596	-0.0005	$+0.395$	$+0.201$	0.0247	0.960
U=10, triv.	padded	0.0316	1.051	1.047	-0.0009	$+0.693$	$+0.356$	0.0435	0.930
U=10, nontriv.	padded	0.01	0.279	0.281	$+0.0003$	$+0.192$	$+0.089$	0.0133	0.978
U=10, nontriv.	padded	0.0178	0.495	0.496	$+0.0005$	$+0.338$	$+0.157$	0.0236	0.962
U=10, nontriv.	padded	0.0316	0.872	0.872	$+0.0009$	$+0.593$	$+0.278$	0.0415	0.933
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.0316	0.018	0.016	$+0.0007$	$+0.004$	-0.021	0.0072	0.991
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.1	0.051	0.050	$+0.0023$	$+0.013$	-0.065	0.0226	0.971
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.178	0.091	0.089	$+0.0041$	$+0.022$	-0.115	0.0399	0.949
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.316	0.157	0.157	$+0.0074$	$+0.039$	-0.203	0.0698	0.912
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.562	0.275	0.275	$+0.0131$	$+0.067$	-0.355	0.1207	0.849
U=0, triv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	1	0.475	0.478	$+0.0234$	$+0.114$	-0.615	0.2040	0.747
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.0178	0.044	0.044	-0.0066	$+0.059$	-0.008	0.0349	0.962
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.0316	0.074	0.075	-0.0119	$+0.102$	-0.015	0.0612	0.934
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.0562	0.125	0.123	-0.0213	$+0.175$	-0.031	0.1061	0.885
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.1	0.187	0.186	-0.0384	$+0.290$	-0.066	0.1805	0.806
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.178	0.245	0.242	-0.0700	$+0.458$	-0.146	0.2970	0.683
U=0, nontriv.	$L_{\mathrm{cut}}^2 = 10^{-5}$	0.316	0.201	0.201	-0.1303	$+0.663$	-0.331	0.4630	0.511
U=0, nontriv.	$T = 100$	0.01	0.186	0.184	$+0.0002$	$+0.184$	$+0.000$	0.0031	0.996
U=0, nontriv.	$T = 100$	0.0178	0.325	0.327	$+0.0003$	$+0.326$	$+0.000$	0.0056	0.993
U=0, nontriv.	$T = 100$	0.0316	0.580	0.579	$+0.0006$	$+0.578$	$+0.001$	0.0099	0.988
U=0, nontriv.	$T = 100$	0.0562	1.022	1.024	$+0.0010$	$+1.021$	$+0.001$	0.0175	0.980
U=10, triv.	angles $+1 \cdot 10^{-13}$	0.01	0.272	0.269	$+0.0000$	$+0.222$	$+0.047$	0.0134	0.977
U=10, triv.	angles $+2 \cdot 10^{-13}$	0.01	0.499	0.500	$+0.0004$	$+0.391$	$+0.108$	0.0148	0.975
U=0, triv.	T_1/T_2 only	0.1	0.052	0.051	-0.0008	$+0.047$	$+0.005$	0.0124	0.986
U=0, nontriv.	T_1/T_2 only	0.1	0.200	0.197	-0.0023	$+0.178$	$+0.021$	0.0336	0.964
U=10, triv.	T_1/T_2 only	0.01	0.206	0.204	-0.0002	$+0.151$	$+0.053$	0.0071	0.987
U=10, nontriv.	T_1/T_2 only	0.01	0.166	0.163	$+0.0002$	$+0.123$	$+0.040$	0.0067	0.988
U=0, triv.	p_1 only	0.1	0.011	0.000	-0.0000	$+0.000$	$+0.000$	0.0001	1.000
U=0, nontriv.	p_1 only	0.1	0.013	0.001	-0.0000	$+0.001$	$+0.000$	0.0001	1.000
U=10, triv.	p_1 only	0.01	0.017	0.002	-0.0000	$+0.002$	$+0.001$	0.0001	1.000
U=10, nontriv.	p_1 only	0.01	0.015	0.002	$+0.0000$	$+0.001$	$+0.000$	0.0001	1.000
U=0, triv.	p_2 only	0.1	0.050	0.048	-0.0007	$+0.042$	$+0.007$	0.0104	0.987
U=0, nontriv.	p_2 only	0.1	0.154	0.152	-0.0011	$+0.127$	$+0.026$	0.0232	0.976
U=10, triv.	p_2 only	0.01	0.190	0.188	-0.0001	$+0.130$	$+0.058$	0.0069	0.990
U=10, nontriv.	p_2 only	0.01	0.151	0.152	$+0.0000$	$+0.106$	$+0.047$	0.0066	0.990
U=0, triv.	readout only	0.1	0.011	0.003	-0.0002	-0.000	$+0.004$	0.0000	0.999
U=0, nontriv.	readout only	0.1	0.012	0.009	-0.0002	-0.001	$+0.010$	0.0000	0.999
U=10, triv.	readout only	0.01	0.014	0.001	-0.0000	$+0.000$	$+0.001$	0.0000	1.000
U=10, nontriv.	readout only	0.01	0.018	0.001	$+0.0000$	-0.001	$+0.002$	0.0000	1.000
U=0, nontriv.	CNOTs folded $\times 3$	0.1	0.987	0.986	-0.0108	$+0.852$	$+0.145$	0.1581	0.836
U=10, nontriv.	CNOTs folded $\times 3$	0.01	0.927	0.928	$+0.0008$	$+0.668$	$+0.259$	0.0392	0.937

Table 9: Shot dependence. Threshold s^* at three shot counts (same rule), and the shot floor S^* (fewest shots per circuit with RMSE $\leq \varepsilon$) without gate noise ($s = 0$) and at $s = 0.01$.

configuration	$s^*(S{=}10^3)$	$s^*(S{=}10^4)$	$s^*(S{=}10^5)$	$S^*(s{=}0)$	$S^*(s{=}0.01)$
U=0, triv.	0.321	0.309	0.315	< 10	13.3
U=0, nontriv.	0.0859	0.0859	0.0859	< 10	< 10
U=10, triv.	0.00736	0.00762	0.00765	18.5	$> 10^6$
U=10, nontriv.	0.0101	0.00974	0.00982	23.1	300

Table 10: Fraction of seeds ($S = 10^4$) for which *both* topological phases are classified correctly ($d < \pi/2$), versus noise scale s.

s	0	0.001	0.00178	0.00316	0.00562	0.01	0.0178	0.0316	0.0562	0.1	0.178	0.316	0.562	1	1.78
U=0	1.00	–	–	–	–	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	0.00
U=10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00	0.00	1.00	0.00	0.91	0.00	–

Table 11: Systematic over-rotation $\theta \to (1+\kappa)\theta$ (no incoherent noise, infinite shots): leg biases, total circular deviation, and the geometric-leg bias in units of $\kappa|\varphi_{G,2}|$.

configuration	κ	$\Delta\varphi_{\mathrm{qc}}$	$\Delta\varphi_{G,1}$	$\Delta\varphi_{G,2}$	d^{∞}	$\lvert\Delta\varphi_{G,2}\rvert/(\kappa\lvert\varphi_{G,2}\rvert)$
U=0, triv.	0.001	$+0.0006$	$+0.0009$	$+0.0017$	0.0031	0.55
U=0, triv.	0.003	$+0.0016$	$+0.0031$	$+0.0052$	0.0098	0.56
U=0, triv.	0.01	$+0.0053$	$+0.0133$	$+0.0176$	0.0361	0.56
U=0, triv.	0.03	$+0.0144$	$+0.0652$	$+0.0545$	0.1341	0.58
U=0, nontriv.	0.001	-0.0065	$+0.0055$	$+0.0026$	0.0016	4.18
U=0, nontriv.	0.003	-0.0195	$+0.0184$	$+0.0083$	0.0072	4.41
U=0, nontriv.	0.01	-0.0655	$+0.0849$	$+0.0325$	0.0519	5.18
U=0, nontriv.	0.03	-0.2021	$+0.4539$	$+0.1387$	0.3904	7.36
U=10, triv.	0.001	$+0.0041$	$+0.0012$	$+0.0117$	0.0170	0.74
U=10, triv.	0.003	$+0.0124$	$+0.0106$	$+0.0356$	0.0586	0.76
U=10, triv.	0.01	$+0.0411$	$+0.1173$	$+0.1257$	0.2841	0.80
U=10, triv.	0.03	$+0.1205$	$+1.0337$	$+0.4346$	1.5887	0.92
U=10, nontriv.	0.001	-0.0003	-0.0002	$+0.0175$	0.0171	1.07
U=10, nontriv.	0.003	-0.0008	$+0.0045$	$+0.0531$	0.0568	1.09
U=10, nontriv.	0.01	-0.0029	$+0.0733$	$+0.1831$	0.2536	1.12
U=10, nontriv.	0.03	-0.0093	$+0.6989$	$+0.5990$	1.2886	1.23